Research Methodology in Computer Science

UNIT I WORKSHEET

Introduction to Research Problem



Course: CSEG3060 | Unit: I | Lecture Allocation: 9 Hours
Designed as an active-learning revision and practice worksheet.

Name: Kushagra Singh **Roll No.:** 590012353

Section: B1[CCVT] **Date:** 15/9/2026

How to use this worksheet

This worksheet is meant to move you from *"I know the definitions"* to *"I can formulate and defend a research problem."* Work through the activities in order. Use your own Computer Science topic whenever a question asks you to design or formulate something.

Unit I Coverage Checklist

- ☑ Meaning and significance of research
- ☑ Objectives of research
- ☑ Key elements of research
- ☑ Sources of research problems
- ☑ Identification of research problems
- ☑ Good research problem criteria
- ☑ Common selection errors
- ☑ Avoiding selection errors
- ☑ Scope and boundaries
- ☑ Research objectives
- ☑ SMART objectives
- ☑ Research approaches
- ☑ Theoretical vs. empirical research
- ☑ Data types and sources
- ☑ Data collection tools
- ☑ Data collection guidelines
- ☑ Data analysis
- ☑ Interpretation of results
- ☑ Research instrumentation
- ☑ Calibration and repeatability
- ☑ Integrated case-based application

1 Part A — Research: What, Why and How?

A1. Research in one sentence

Research is a **systematic and scientific investigation** used to discover new facts, verify existing knowledge, analyze relationships, explain phenomena, and answer questions using evidence.
Key elements from class:

Problem Identification → Data Collection & Analysis → Hypothesis Formation → Result Interpretation

Activity 1: Build the research chain

Put the following in a sensible order:

Result Interpretation	Problem Identification	Data Collection/Analysis	Hypothesis Formation
4	1	2	3

Why is this order useful in a CS research project? **[2 marks]**

Each stage feeds the next: a clear problem guides data collection, analysis enables a testable hypothesis, and interpretation turns results into an answer to the problem.

Activity 2: Why does research matter?

Give one CS example for each:

Technology/knowledge advancement	Transformer models advancing natural-language processing.
Industrial innovation	Recommender systems boosting e-commerce sales.
Academic growth	New graph algorithms published and cited in later work.
Policy or societal decision-making	Predictive models guiding public-health resource planning.
Computer Science example (AI/algorithms/HCI/etc.)	ML-based intrusion detection with fewer false positives.

Activity 3: Research objectives

Complete the table.

Objective of research	Your CS example
Find answers using scientific methods	Measure how CNN depth affects image-classification accuracy.
Explain phenomena / establish relationships	Study how dataset size relates to model overfitting.
Test or validate an existing theory	Empirically test whether dropout reduces overfitting.

Think like a researcher

A student says: *"I want to build an AI app because AI is trending."* Is this already a research problem? Explain what is missing.

No. It states a trend-driven wish, not a problem. It lacks a specific gap, a research question, defined scope, measurable objectives, and success criteria - it says what to build, not what to investigate.

2 Part B — Types and Approaches of Research

Research approaches covered in Unit I

Exploratory
discover patterns/ideas

Descriptive
describe facts/characteristics

Analytical
analyze information

Theoretical
models, logic, proofs

Empirical
observation/experiment

The class material also introduced **fundamental** and **applied** research as broad research types. Choose the approach according to the research question and the kind of evidence required.

Activity 4: Match the approach

Write E (Exploratory), D (Descriptive), A (Analytical), T (Theoretical), or EM (Empirical).

1. Investigating an emerging problem where very little is known: E
2. Measuring average response time of five search algorithms: EM
3. Developing a formal model for a new computational system: T
4. Summarizing user characteristics from a usability survey: D
5. Comparing existing datasets and identifying relationships among variables: A

Activity 5: Theoretical or empirical?

For each scenario, decide the dominant approach and justify it in one sentence.

Scenario	Type	Why?
Prove a property of an algorithm mathematically	Theoretical	Uses formal logic and mathematical proof, not data.
Benchmark an intrusion-detection model on a dataset	Empirical	Conclusions drawn from measured experimental results.
Develop a formal model of a distributed protocol	Theoretical	Builds an abstract model rather than collecting data.
Interview developers to discover pain-points in an emerging tool	Empirical	Gathers first-hand qualitative data via interviews.

3 Part C — Finding and Formulating a Research Problem

C1. Where do research problems come from?

Important sources highlighted in class include:

- Literature reviews and research gaps.
- Industrial problems and real-world needs.
- Emerging trends and technologies.
- Academic coursework and past projects.
- Surveys of recent publications.
- Patent and research databases.
- Industry white papers.
- Competitive research proposals.

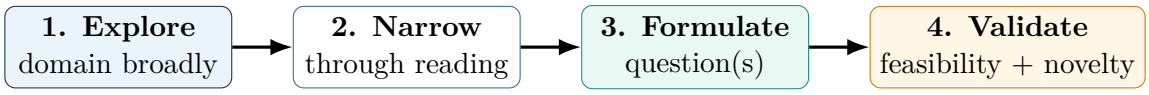
Activity 6: Research-problem detective

Pick a CS domain: Network security / intrusion detection

List **four possible sources** from which a research problem in this domain could emerge:

1. Literature gaps found in ML-IDS survey papers.
2. High false-positive rates reported by industry SOC teams.
3. Emerging attack types from IoT and 5G deployments.
4. Public benchmark datasets such as CICIDS2017.

C2. Identification workflow



Activity 7: Apply the four-step workflow

Topic/domain: Machine learning for network intrusion detection

1. Explore broadly: What is the larger area?

Cybersecurity and network-traffic analysis.

2. Narrow through reading: What specific sub-area or pain-point interests you?

Reducing false positives in ML-based intrusion detection.

3. Formulate question(s): Write one research question.

Can feature selection lower the false-positive rate of an ML NIDS without hurting recall?

4. Validate: What would you check for feasibility and novelty?

Dataset availability, compute cost, and how it differs from existing IDS studies.

Quick Pair Challenge: Problem or Topic?

Classify each statement as **Topic**, **Weak Problem**, or **Research Problem**.

1. "Artificial Intelligence in Smart Cities." Topic
2. "AI is interesting." Weak Problem
3. "How can reinforcement learning optimize traffic signals in smart cities?" Research Problem
4. "Cybersecurity." Topic
5. Your own statement: How can ML cut false positives in NIDS?

C3. Characteristics of a good research problem

A strong problem should be **clear and concise, researchable and feasible, relevant and original**, and should offer scope for **data collection and analysis**. The syllabus also emphasizes essential attributes and avoiding errors in selection. **[5 marks]**

Criterion	Evidence that YOUR problem satisfies it
Clear and concise	States one clear goal: cut false positives in an ML NIDS.
Researchable	Testable on public IDS datasets with standard metrics.
Feasible	Uses available datasets, ML libraries, and a single GPU.
Relevant	False alarms overwhelm analysts, so reducing them helps SOCs.
Original / non-trivial	Targets FP reduction, less studied than raw accuracy.
Allows data collection and analysis	Yields precision, recall, and FP-rate for quantitative analysis.

4 Part D — What Can Go Wrong? Scope and Boundaries

Common errors in selecting a research problem

1. Ambiguity / vagueness
The problem does not clearly state what is being investigated.
2. Too broad
The study becomes unrealistic or unmanageable.
3. Too narrow
There may be insufficient room for meaningful investigation.
4. Lack of relevance
The problem has weak academic, practical, or societal value.
5. Lack of feasibility
Resources, data, skills, time, or infrastructure are unavailable.
6. Ignoring prior art
Existing research may already answer the question.

Activity 8: Error clinic

For each weak problem, identify the main error and rewrite it.

1. "Improve AI everywhere."
   Error: Too broad / vague - unmanageable with no clear focus.
   Improved version:
   Improve CNN inference speed on edge devices for real-time image classification.

2. "Study all cybersecurity threats affecting every organization worldwide."
   Error: Too broad and infeasible in scope.
   Improved version:
   Study phishing-URL detection accuracy for a university email system over one semester.

3. "Build a system that detects every possible disease with 100% accuracy in two days."
   Error: Unrealistic and infeasible target and timeline.
   Improved version:
   Develop an ML model to grade diabetic-retinopathy severity from retinal images with high recall.

How to avoid selection errors

- Perform a literature review.
- Consult experts or guides.
- Conduct pilot studies where appropriate.
- Align the problem with objectives and available resources.

D2. Scope = the boundary of your research

Scope defines the **limits and boundaries** of the study and prevents deviation from the objectives. Typical boundaries include:

Time frame | **Geographic** location | **Target population** or dataset | **Tools /** technologies

Activity 9: Draw the boundary

Research problem: *"How can machine learning improve network intrusion detection?"*

Fill in realistic boundaries:

Time frame	3 months (one project cycle).
Geographic / deployment context	Enterprise LAN; offline dataset evaluation.
Target dataset / population	CICIDS2017 network-flow dataset.
Tools / algorithms	Python, scikit-learn; Random Forest and XGBoost.
What is explicitly OUT of scope?	Live deployment, encrypted-payload inspection, zero-day attacks.

Activity 10: Scope test

A student's project is titled *"AI for Smart Cities."* Give three reasons why this scope is likely too broad, then convert it into a narrower study.

Too broad because: (1) 'AI' covers many techniques; (2) 'smart cities' spans traffic, energy, water and safety; (3) no dataset, metric, or timeframe is fixed. Narrower study: 'Using reinforcement learning to reduce average vehicle waiting time at signalized intersections in a simulated city grid.'

5 Part E — Research Objectives and SMART Thinking

E1. Why objectives matter

Research objectives:

- set the direction and purpose of research;
- help define scope and methodology; and
- provide focus and coherence.

A good objective is not just an activity such as "study AI." It states **what will be achieved or evaluated**.



Activity 11: SMART objective laboratory

Research topic/problem: ML-based network intrusion detection with fewer false positives

Write one objective, then audit it.

Specific	What exactly will be done? Build and tune an ML classifier that flags malicious network flows.
Measurable	What variable, metric, or evidence will show achievement? Reduce the false-positive rate below 2% while keeping recall at or above 95%.
Achievable	Why is it realistic with available resources? Public datasets and standard ML libraries make it feasible on one GPU.
Relevant	How does it address the research problem? Directly tackles alert fatigue, the core problem for SOC analysts.
Time-bound	By when / over what period? Completed within a 12-week semester project.

Final SMART objective:

Within 12 weeks, develop an ML NIDS on CICIDS2017 that lowers the false-positive rate below 2% while maintaining recall of at least 95%.

Peer Review — 60-second test

Exchange your objective with a partner. Your partner must answer:

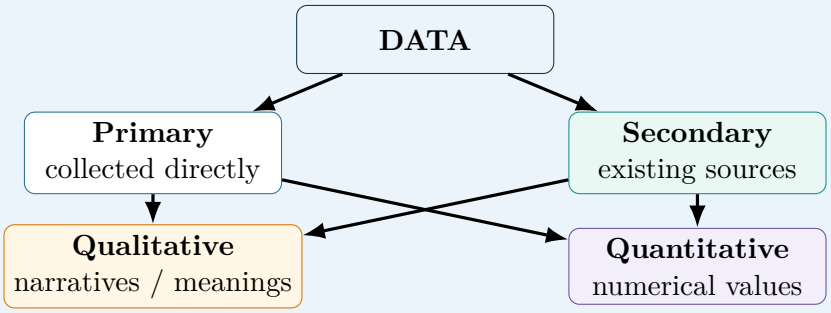
1. Can I identify exactly what is being investigated?
2. Can I identify how success will be measured?
3. Is the target realistic?
4. Does it directly connect to the problem?
5. Is a time boundary visible?

Partner's one improvement suggestion:

Specify the exact dataset and baseline model used for comparison.

6 Part F — Data Fundamentals

F1. Two useful ways to classify data



Examples discussed in class include primary interviews, sensor readings, and usage logs; secondary GitHub/Kaggle datasets, IEEE articles, and government data.

Activity 12: Classify the data

Mark each as Primary/Secondary and Qualitative/Quantitative where applicable.

Data source	Primary/Secondary	Qualitative/Quantitative
Interview transcripts	Primary	Qualitative
Sensor temperature readings	Primary	Quantitative
Kaggle dataset downloaded for analysis	Secondary	Quantitative
System CPU utilization logs collected during your experiment	Primary	Quantitative
User comments about usability	Primary	Qualitative
IEEE article used as an existing source	Secondary	Qualitative

F2. Choosing data

For your own research problem:

1. What data do you need?

   Labeled network-flow records (benign vs. attack).

2. Why is this data appropriate?

   They directly support training and evaluating an intrusion classifier.

3. Primary, secondary, or a combination?

   Combination: public datasets plus self-captured lab traffic.

4. Qualitative, quantitative, or a combination?

   Mainly quantitative (numeric flow features and metrics).

7 Part G — Data Collection Tools and Guidelines

G1. Match the tool to the evidence

The class material highlighted:

- **Questionnaires / online forms** — structured responses from participants.
- **Observation logs / experiments** — record observed behavior or experimental events.
- **System monitoring tools** — capture system/network performance.
- **Web scraping** — collect information from web sources using appropriate tools.

Examples shown in the class material include Google Forms, Typeform, Wireshark, PerfMon, BeautifulSoup, and Selenium.

Activity 13: Tool selection game

You need to collect the following evidence. Select the most suitable tool and justify.

Evidence required	Tool / method	Why?
Students' satisfaction with an app	Questionnaire (Google Forms)	Collects structured self-reported feedback at scale.
Network packet behavior during an experiment	Wireshark	Captures and inspects live packet-level traffic.
CPU/memory usage of a program	PerfMon / system monitor	Logs CPU and memory use during runs.
Behavior of users interacting with a prototype	Observation log / usability test	Records real interaction behaviour directly.
Public web data needed for a study	Web scraping (BeautifulSoup)	Automates collection of public web data.

Activity 14: Design a mini data-collection instrument

Choose one: ☐questionnaire ☐observation log ☐experiment log ☑system-monitoring plan.

Research problem: ML-based network intrusion detection

What will be measured/recorded?	Per-flow features and predicted labels (attack/benign).
Who/what is the source?	Network flows from CICIDS2017 and a lab testbed.
When/how often?	Continuously during each experiment run.
What tool will be used?	Wireshark + CICFlowMeter; Python logging.
How will consistency be maintained?	Fixed capture settings and one shared feature pipeline.

8 Part H — From Data to Meaning: Analysis and Interpretation

H1. Do not confuse these stages

Collect raw evidence → **Analyze** organize/compare → **Interpret** explain meaning → **Conclude** answer problem

Analysis asks: "What patterns, differences, relationships, or measurements are in the data?"
Interpretation asks: "What do those findings mean for my research problem and objectives?"

Activity 15: Read the result, then interpret it

Suppose an experiment gives:

Model	Accuracy	Inference time (ms)
A	92%	20
B	95%	70
C	94%	25

1. Which model has the highest accuracy?

   Model B (95%).
2. Which model is fastest?

   Model A (20 ms).
3. If the system is real-time and accuracy must be at least 93%, which model would you investigate first? Why?

   Model C: it meets the 93% bar (94%) with near-fastest latency (25 ms) - best speed/accuracy balance.
4. Is "B is best" a valid conclusion without considering the research objective? Explain.

   No - B's 70 ms latency fails real-time needs; 'best' depends on the objective, not accuracy alone.

Activity 16: Interpretation vs. description

Complete each sentence with either **description** or **interpretation**.

1. "Model B achieved 95% accuracy." description
2. "The higher accuracy may be useful when accuracy is the primary objective, but its latency may limit real-time deployment." interpretation
3. "The system processed 10,000 requests." description
4. "The observed throughput suggests the system can support the target workload." interpretation

Researcher's caution

A graph can show a pattern, but a pattern is not automatically an explanation. Write one question you would ask before claiming *why* a result occurred.

Could a confounding factor (e.g., dataset imbalance or hardware) explain the result rather than the model itself?

9 Part I — Research Instrumentation

I1. Instrumentation in computing research

Research instrumentation refers to the **tools, hardware, and software used for data collection and analysis**. Class examples include:

- sensors;
- simulators;
- benchmarking suites;
- custom tools for robotics, networks, and AI.

Two important quality ideas emphasized in the class material are **calibration** and **repeatability**.

Activity 17: Instrument selection matrix

For your chosen research problem, identify one instrument at each level.

Level	Possible instrument	What evidence will it produce?
Hardware / sensor	Network TAP / NIC in promiscuous mode	Raw packet captures from the link.
Software / simulator	NS-3 / Mininet traffic simulator	Reproducible synthetic attack traffic.
Benchmark / measurement tool	scikit-learn metrics + CICFlowMeter	Precision, recall, FP-rate, flow features.
Custom tool (if needed)	Python feature-extraction & logging script	Labeled feature vectors for training.

Activity 18: Calibration and repeatability

Why should a sensor, measurement setup, benchmark, or custom instrument be calibrated?

So its measurements are accurate and unbiased, giving trustworthy, comparable data.

What could happen if the same experiment cannot be repeated consistently?

Results become unreliable and unverifiable, so conclusions cannot be trusted or reproduced.

10 Part J — The Research Problem Builder

Your 10-minute research design

Choose a real Computer Science problem you care about. Complete every box.

Step 1 — Domain

Network security (intrusion detection).

Step 2 — Real-world pain-point

IDS raise too many false alarms, causing analyst alert fatigue.

Step 3 — Research problem statement

ML-based NIDS often trade fewer false positives for lower detection.

Step 4 — Research question

Can feature selection cut false positives without reducing recall?

Step 5 — Why is it worth investigating?

Fewer false alarms let analysts focus on genuine threats.

Step 6 — Research approach

☐Exploratory ☐Descriptive ☐Analytical ☐Theoretical ☑Empirical ☐Fundamental ☑Applied

Step 7 — Scope boundaries

Time: 3 months Dataset/population: CICIDS2017

Tools/technology: Python, scikit-learn Out of scope: Live deployment, zero-days

Step 8 — Three SMART objectives

O1: Build a baseline Random-Forest classifier on CICIDS2017.

O2: Apply feature selection to cut the false-positive rate below 2%.

O3: Compare recall and FP-rate against the baseline within 12 weeks.

Step 9 — Data plan

Data type(s): Quantitative flow features Source(s): CICIDS2017 + lab capture

Collection tool(s): CICFlowMeter, Wireshark Instrumentation: Python ML pipeline

Step 10 — Analysis and interpretation

What will you measure/analyze? How will the findings answer the research question?

Measure precision, recall, and FP-rate; a lower FP-rate at fixed recall shows the method answers the question.

11 Part K — Case Study: AI in Smart Cities

Case Study — Refine the vague idea

A student wants to research:

"Artificial Intelligence in Smart Cities."

Your task is to transform this broad idea into a defensible Unit I research design.

K1. Narrow the problem

Broad area: **AI + Smart Cities**

Choose one specific application: ☑traffic ☐waste ☐energy ☐water ☐safety ☐other: _____

Specific pain-point:

Fixed-timer signals cause congestion at variable-demand intersections.

Refined research problem:

Can reinforcement learning reduce average vehicle waiting time at signalized intersections versus fixed-time control?

K2. Define the approach

Dominant approach: Empirical (simulation-based)

Why is it theoretical, empirical, exploratory, descriptive, analytical, or a combination?

Empirical: it is tested by running RL on a traffic simulator and measuring outcomes.

K3. Three SMART objectives

1. Model a 4-intersection grid in SUMO with realistic demand.
2. Train a DQN signal controller and measure waiting time.
3. Cut mean waiting time by at least 15% vs. fixed-time within 10 weeks.

K4. Boundaries

Timeframe	10 weeks.
Target dataset / population	Simulated traffic flows (SUMO scenarios).
Geographic/deployment context	One simulated urban grid (offline).
Tools/technologies	SUMO, Python, TensorFlow (DQN).
Out of scope	Real-world deployment, pedestrians, multi-city networks.

12 Part L — Case-Based Thinking

Case 1: "Build a better chatbot"

A student proposes: *"I will make a chatbot that is better than ChatGPT."*
Identify at least **four** problems with this research proposal at the Unit I level.

1. 'Better than ChatGPT' is vague and not measurable.
2. Scope is far too broad and infeasible for a student project.
3. No specific task, dataset, or evaluation metric is defined.
4. Ignores prior art and available resources.

Rewrite it as a focused research problem:

Can a fine-tuned small LLM improve FAQ-answering accuracy for a university helpdesk versus a retrieval baseline?

Case 2: Sensor-based smart agriculture

A team wants to predict irrigation needs using soil-moisture sensors.
Identify:

1. One possible research problem.
   Can soil-moisture data predict a crop's daily irrigation need?
2. Two data sources.
   Soil-moisture sensors and local weather records.
3. One primary data source.
   Field soil-moisture sensor readings.
4. One secondary data source.
   Meteorological-department rainfall data.
5. One data-collection tool.
   IoT sensor with a data logger.
6. One research instrument.
   Calibrated soil-moisture sensor.
7. One measurable objective.
   Predict irrigation need within +/-10% over one season.
8. One scope boundary.
   One crop type on a single farm plot.

Case 3: The "perfect accuracy" trap

A researcher claims: *"My model is successful because it gives 99.9% accuracy."*
Write three questions you would ask before accepting the claim:

1. On what dataset and train/test split was 99.9% measured?
2. Is the data balanced, or does accuracy hide poor recall?
3. Does it generalize to unseen, real-world data?

13 Part M — Unit I Rapid-Fire Quiz

Choose the best answer.

1. A research problem should primarily be:

   (a) fashionable
   (c) extremely broad
   (b) clear and researchable
   (d) impossible to test

2. Which is a source of research problems?

   (a) literature gaps
   (c) avoiding prior work
   (b) random guessing only
   (d) unrelated facts

3. "Measure the average latency of Algorithm X on Dataset Y within four weeks" is closest to:

   (a) vague topic
   (c) literature review
   (b) SMART objective
   (d) research instrument

4. Existing data from Kaggle is generally:

   (a) primary
   (c) always qualitative
   (b) secondary
   (d) always experimental

5. "User comments about usability" are primarily:

   (a) qualitative
   (c) hardware
   (b) quantitative
   (d) benchmark

6. Which is most directly related to system monitoring?

   (a) PerfMon
   (c) patent claim
   (b) title page
   (d) abstract

7. A study based mainly on observation and experiments is:

   (a) empirical
   (c) descriptive writing
   (b) purely theoretical
   (d) scope definition

8. Which is NOT a good scope boundary?

   (a) dataset
   (c) tools
   (b) timeframe
   (d) "everything"

9. Interpretation primarily asks:

   (a) What does the result mean?
   (c) Who printed the report?
   (b) What font was used?
   (d) What is the page number?

10. Calibration and repeatability are especially important for:

(a) research instrumentation

(b) research titles

(c) roll numbers

(d) acknowledgements

Rapid-fire: One-word / one-line

1. Define research in your own words:

   Systematic, evidence-based investigation to answer questions or verify knowledge.

2. One difference between a topic and a research problem:

   A topic is a broad area; a research problem is a specific, answerable question.

3. One source of a research problem:

   Literature gaps.

4. One common selection error:

   Making the scope too broad.

5. Expand SMART:

   Specific, Measurable, Achievable, Relevant, Time-bound.

6. One primary-data example in CS:

   CPU-utilization logs from your own experiment.

7. One secondary-data example in CS:

   A Kaggle or IEEE dataset.

8. One data collection tool:

   Wireshark.

9. One research instrument:

   A benchmarking suite.

10. Difference between analysis and interpretation:

    Analysis finds patterns; interpretation explains what they mean.

14 Part N — Connect the Dots

Match Column A with Column B.

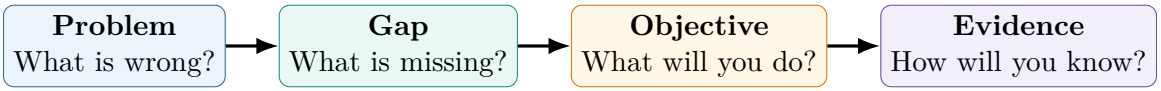
A	Concept	B	Meaning / Example
1	Exploratory	A	Collected directly
2	Primary data	B	Clear, measurable, realistic, relevant, time-bound
3	Scope	C	Discover patterns / new ideas
4	SMART	D	Limits and boundaries of a study
5	Empirical	E	Observation / experiment based
6	Instrumentation	F	Tools/hardware/software for collection and analysis
7	Secondary data	G	Existing datasets/articles/logs
8	Interpretation	H	Explain what findings mean for the problem

Answers: 1-C, 2-A, 3-D, 4-B, 5-E, 6-F, 7-G, 8-H.

15 Part O — Viva / Oral Defense Practice

Imagine your teacher asks: *"Defend your research problem in 60 seconds."*

Use the structure:



Your 60-second defense notes:
Problem: Gap: Objective: Evidence/data:

Problem: SOC analysts are flooded by false IDS alerts. Gap: Little work targets false-positive reduction. Objective: Cut the FP-rate below 2% while keeping recall at or above 95%. Evidence: precision, recall, and FP-rate on CICIDS2017.

16 Part P — Unit I Master Check

Can I do it without looking at the PPT?

Rate yourself honestly: **1 = not yet, 2 = partly, 3 = confidently**.

Skill	1	2	3
Explain what research is and why it matters			✓
Identify the key elements of research			✓
Generate sources of research problems			✓
Distinguish a topic from a research problem			✓
Evaluate a research problem using quality criteria			✓
Identify common selection errors			✓
Define scope and boundaries			✓
Write SMART objectives			✓
Choose an appropriate research approach			✓
Distinguish theoretical and empirical research			✓
Classify primary/secondary data			✓
Classify qualitative/quantitative data			✓
Select a suitable collection tool			✓
Explain analysis vs interpretation			✓
Select suitable research instrumentation			✓
Explain why calibration and repeatability matter			✓

My one-minute takeaway

The most important idea I learned in Unit I is:

A good research problem must be specific, feasible, and measurable - not just a trendy topic.

The concept I still need to revise is:

Writing fully SMART objectives quickly.

One research problem I can now formulate is:

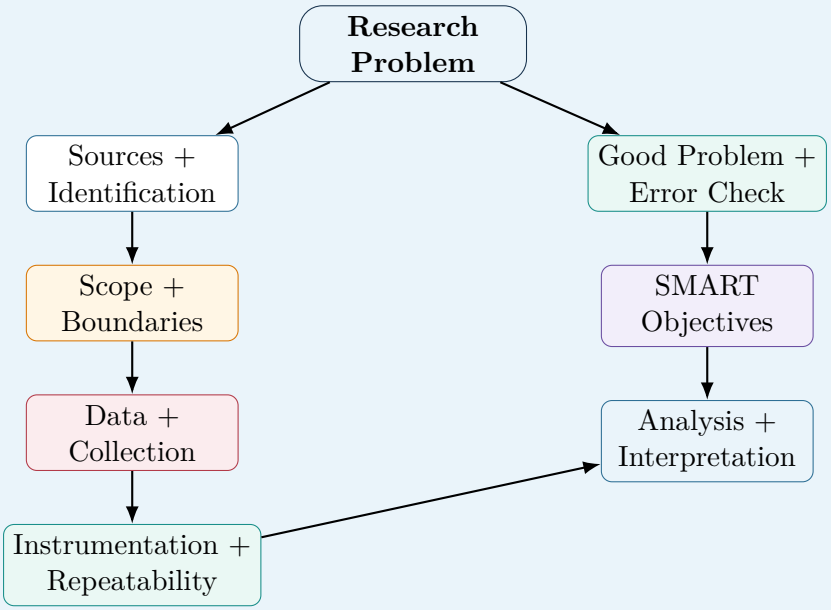
Reducing false positives in ML-based network intrusion detection.

Exit Ticket: Researcher Mindset

Complete the sentence:
"A good researcher does not start by asking which tool to build *; a good researcher first asks* what problem to solve *."*
Explain your answer in one or two lines:

Tools are only a means; research must be driven by a clear, worthwhile problem and question first.

Unit I in one visual



End of Unit I Worksheet — Introduction to Research Problem

Research Methodology in Computer Science

WORKSHEET–02: UNIT II

Research Methodology in Computer Science

Dialectic, Arguments, Proofs, Deduction, Induction, Models and Evaluation

Name: Kushagra Singh **Roll No.:** 590012353

Section: B1[CCVT] **Date:** 15/9/2026

LEARN & REMEMBER

Purpose of this worksheet

This worksheet helps you understand and apply the major concepts of Unit II. Read the short explanation, complete the examples, and then attempt the practice questions.

CLASSROOM ACTIVITY

Unit II in one line

Claim → Argument → Challenge → Proof/Evidence → Model → Evaluation

UNIT II — LEARNING MAP

Tick each topic after completing it.

#	Topic	Done	Confidence
1	Dialectic of Research in CS	☑	1–5 5
2	Models of Argument	☑	1–5 4
3	Proof Methods	☑	1–5 5
4	Proof by Demonstration	☑	1–5 4
5	Empirical Proofs	☑	1–5 5
6	Mathematical Proofs	☑	1–5 4
7	Deduction and Induction	☑	1–5 4
8	Theoretical Models	☑	1–5 5
9	Algorithmic Approaches	☑	1–5 4
10	Software Engineering Approaches	☑	1–5 4
11	Mathematical Modelling	☑	1–5 5
12	Performance Evaluation	☑	1–5 4

PRACTICE ZONE

Quick warm-up

1. General → specific reasoning is called:

   Deduction

2. Specific observations → general rule is called:

   Induction

3. Reasoning toward the "best explanation" is called:

   Abduction

4. A claim becomes stronger when it is:

   tested against counter-arguments and supporting evidence

PART A — DIALECTIC OF RESEARCH

Using claims, counterclaims and evidence to refine research

LEARN & REMEMBER

Dialectic is structured reasoning through opposing or different views.
In Computer Science research, the process can be understood as:

$$\boxed{\text{Claim} \rightarrow \text{Challenge} \rightarrow \text{Evidence} \rightarrow \text{Refinement}}$$

It helps researchers:

- identify weaknesses in an idea,
- consider alternative viewpoints,
- reduce premature conclusions,
- refine vague research ideas,
- strengthen research arguments.

A1. Dialectical Process

1. Identify a claim or hypothesis.
2. Challenge it using counter-arguments or experiments.
3. Examine evidence and logic.
4. Refine the claim.
5. Arrive at a stronger explanation or solution.

PRACTICE ZONE

Practice
Claim:
"Greedy algorithms always produce optimal results."

1. What is the claim?
   Greedy algorithms always give the optimal solution.
2. Give a counterexample or situation where it may fail.
   0/1 knapsack: greedy by value/weight is not optimal.
3. Write a more accurate version of the claim.
   Greedy is optimal only when the problem has the greedy-choice property.
4. What evidence could support the revised claim?
   A proof of the greedy-choice property, plus counterexamples where it fails.

A2. Dialectic and Problem Formulation

Choose one:

AI in Education Smart Traffic Robotic Farming (IoT Security)

Initial idea	Detect intrusions in IoT networks using machine learning.
Possible claim	ML models can accurately flag IoT attacks in real time.
Counter-argument	IoT devices are resource-limited and traffic data is imbalanced.
Evidence required	Detection accuracy and latency on IoT-traffic datasets.
Refined research problem	Can a lightweight ML IDS detect IoT attacks with few false positives on-device?

PART B — MODELS OF ARGUMENT

Deductive, inductive and abductive reasoning

LEARN & REMEMBER

There are three important reasoning models:

	Deductive	Inductive	Abductive
Direction	General → Specific	Specific → General	Observation → Explanation
Conclusion	Certain if premises are true	Probable	Plausible
Typical CS use	Proof	ML / data analysis	Debugging

B1. Deductive Reasoning

General rule → specific case → conclusion.

Example

All Dijkstra results are optimal when its assumptions are satisfied.

Graph G satisfies those assumptions.

Therefore, the shortest-path result for G is optimal.

PRACTICE ZONE

Create your own deductive argument.

General rule:

All comparison-based sorts need >= n log n comparisons (worst case).

Specific case:

Merge sort is a comparison-based sort.

Conclusion:

So merge sort needs >= n log n comparisons in the worst case.

B2. Inductive Reasoning

Specific observations are used to form a general rule.

Example

A new sorting algorithm performs better than Quicksort on 100 test cases.

Possible inductive conclusion:

The new algorithm may perform better in general.

The conclusion is **not guaranteed**.

PRACTICE ZONE

A new ML model performs better on 8 datasets.

1. What are the observations?
   The model beat the baseline on all 8 datasets tested.
2. What general conclusion might be proposed?
   The model may generally outperform the baseline.
3. Why is the conclusion not certain?
   Only 8 datasets were tested; it may not generalize.

B3. Abductive Reasoning

Abduction searches for the most plausible explanation.

Example

Observation: A program crashes randomly.

Possible explanation: Memory leak.

PRACTICE ZONE

Observation: A web application suddenly becomes very slow.
Write three possible explanations.

1. Slow or unindexed database queries.
2. A server memory leak or CPU overload.
3. Network congestion or a sudden traffic spike.

Which explanation would you investigate first?
Check server resource metrics (CPU and memory) first.

PART C — PROOF METHODS

Demonstration, empirical and mathematical approaches

LEARN & REMEMBER

A proof or validation method provides evidence that a research claim is correct, feasible, reliable, or supported.

Method	Basis	Typical use	Main limitation
Demonstration	Examples / prototype	Early validation	Limited cases
Empirical	Experiments / data	Performance studies	Context-dependent
Mathematical	Logic / formal proof	Correctness / theory	Requires formal assumptions

C1. Identify the Method

Write D = Demonstration, E = Empirical, M = Mathematical.

1. Running a prototype on five sample inputs. **D**
2. Benchmarking an ML model on multiple datasets. **E**
3. Proving algorithm correctness using formal logic. **M**
4. Testing network latency under different workloads. **E**
5. Proving a theorem about computational complexity. **M**

C2. Choose the Validation Method

Research Claim	Method	Why?
A sorting algorithm is correct	M	Correctness needs a formal proof.
A new ML model performs better	E	Shown by experiments on datasets.
A robot prototype can follow a path	D	Demonstrated on a working prototype.
A cryptographic construction is secure under assumptions	M	Proved under formal assumptions.

PART D — PROOF BY DEMONSTRATION

Examples and prototypes as early evidence

LEARN & REMEMBER

Proof by demonstration means showing that something works using examples, test cases, or a prototype.
It is:

- easy to understand,
- useful for early research,
- useful for prototypes,
- useful for teaching,
- but not a universal proof.

D1. Example

Claim:

"This new sorting algorithm is efficient."

Suppose we run it on five sample arrays and all five executions are successful.

This demonstrates that it works for those tested examples.

It does **not** prove that it is efficient for every possible input.

PRACTICE ZONE

Consider:
"Our chatbot correctly handles all customer queries."
Design:

1. Test case 1: Ask a common FAQ (e.g., 'What are your opening hours?').
2. Test case 2: Ask an order-status query with a valid order ID.
3. Test case 3: Ask an ambiguous or misspelled question.
4. One corner case that might be missed: A multi-intent or out-of-scope query.

D2. Demonstration vs Proof

Complete:

Demonstration proves that the method works on the specific tested cases.

Demonstration cannot prove that it is correct for every possible input.

A better next step could be empirical testing on large data or a formal proof.

PART E — EMPIRICAL PROOFS

Evidence through observation, experimentation and measurement

LEARN & REMEMBER

Empirical validation uses real observations and experimental data.
A typical empirical study involves:

$$\text{Question} \rightarrow \text{Experiment} \rightarrow \text{Measurement} \rightarrow \text{Analysis} \rightarrow \text{Conclusion}$$

Common applications include:

- algorithm benchmarking,
- ML model evaluation,
- network performance,
- usability studies,
- system experiments.

E1. Example

Claim:

"Our new ML model performs better than the baseline."

Possible measurements:

- Accuracy
- Precision
- Recall
- F1-score
- Runtime

PRACTICE ZONE

Design an empirical experiment for:
"A new network protocol reduces latency."

Independent variable: Protocol used (new vs. baseline).

Dependent variable: Average end-to-end latency (ms).

Baseline: The existing standard protocol (e.g., TCP).

Dataset / workload: Simulated traffic at varying loads.

Metric: Mean latency and jitter.

Number of trials: 30 runs per condition.

Possible limitation: Simulation may not reflect real networks.

E2. Advantages and Limitations

Advantages

- Real-world evidence
- Practical performance
- Supports comparison
- Allows iterative improvement

Limitations

- Context-dependent
- Can be expensive
- External factors matter
- May not generalize

PART F — MATHEMATICAL PROOFS

Rigorous reasoning using logic and mathematics

LEARN & REMEMBER

Mathematical proof provides rigorous validation.
Important types:

1. Direct proof
2. Proof by contradiction
3. Proof by induction
4. Proof by contrapositive

Mathematical proof is widely used in algorithm correctness, complexity, cryptography, automata theory and formal systems.

F1. Direct Proof

A direct proof starts from known assumptions and logically derives the required result.

PRACTICE ZONE

Complete the structure.

Statement to prove:
The sum of two even integers is even.

Given / assumptions:
Let a = 2m and b = 2n for integers m, n.

Logical steps:
a + b = 2m + 2n = 2(m + n).

Conclusion:
a + b = 2(m + n), which is even.

F2. Proof by Contradiction

Basic idea:

Assume opposite → derive contradiction → original claim holds

Example in CS:

Impossibility results such as the Halting Problem can use contradiction-style reasoning.

PRACTICE ZONE

Fill the structure.

Claim:

sqrt(2) is irrational.

Opposite assumption:

Assume sqrt(2) = a/b in lowest terms.

Contradiction:

Then a^2 = 2b^2, so a and b are both even - not lowest terms.

Conclusion:

So sqrt(2) cannot be rational; it is irrational.

F3. Proof by Induction

Induction contains:

1. Base case
2. Inductive hypothesis
3. Inductive step
4. Conclusion

Classic result:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

PRACTICE ZONE

Base case:

For $n = 1$:

$$1 = \frac{1(1+1)}{2}$$

Inductive hypothesis:

Assume:

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$

Inductive step:

$$1 + 2 + \cdots + k + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1)$$

Complete:

$$= \frac{(k+1)(\text{ k + 2 })}{2}$$

Conclusion:

By induction, 1 + 2 + ... + n = n(n + 1)/2 for all n >= 1.

PART G — CONTRAPOSITIVE

A useful form of implication-based reasoning

LEARN & REMEMBER

For an implication

$$P \Rightarrow Q$$

the contrapositive is

$$\neg Q \Rightarrow \neg P$$

An implication and its contrapositive are logically equivalent.

PRACTICE ZONE

Write the contrapositive of:
If an algorithm terminates, then it produces an output.
Original statement:
If an algorithm terminates, then it produces an output.

Contrapositive:
If it produces no output, then it does not terminate.

Explain why the two statements are logically related.
They are logically equivalent - true in exactly the same cases.

G1. Identify the Proof Type

Write:

Direct / Contradiction / Induction / Contrapositive

1. Assume the opposite and derive an impossible result. Contradiction
2. Prove the base case and then $k \to k+1$. Induction
3. Prove $\neg Q \to \neg P$ to establish $P \to Q$. Contrapositive
4. Start from known assumptions and derive the result directly. Direct

G2. Applications

Complete the table.

Algorithm correctness	Direct proof or induction.
Complexity analysis	Asymptotic bounds via induction.
Cryptography	Proof by contradiction / reduction.
Automata / formal languages	Induction and contrapositive.

PART H — DEDUCTION AND INDUCTION

Using evidence and logic to build CS knowledge

LEARN & REMEMBER

Deduction

General rule → Specific case

If the premises are true and the logic is valid, the conclusion follows.

Induction

Specific observations → General rule

The conclusion is probable rather than guaranteed.

H1. Compare

	Deduction	Induction	Abduction
Starts with	General rule	Observations	Observation
Ends with	Specific conclusion	Generalization	Best explanation
Certainty	High if premises true	Probable	Plausible

H2. Classify the Following

Write D, I or A.

1. A theorem is applied to a particular graph. D
2. A researcher finds the same trend in 100 experiments and proposes a general rule. I
3. A program crashes and the researcher suspects a memory leak. A
4. A formal proof establishes algorithm correctness. D

PRACTICE ZONE

Explain one answer

Choose any one of the above and explain why it is deductive, inductive or abductive.

Item 2 is inductive: a general rule is proposed from many specific observations.

PART I — THEORETICAL MODELS

Abstract models for understanding computational systems

LEARN & REMEMBER

Theoretical models provide abstract representations of computational systems.
Important examples include:

- Finite Automata
- Pushdown Automata
- Turing Machines
- Propositional and Predicate Logic
- Graphs and Trees
- Boolean Algebra
- Formal Grammars

They help researchers reason about computation without depending completely on a particular physical implementation.

I1. Match the Model

Recognize a formal language: Finite Automaton

Represent a network structure: Graph

Study computability: Turing Machine

Represent Boolean operations: Boolean Algebra

Reason about logical statements: Propositional / Predicate Logic

I2. Turing Machine

A Turing Machine is an abstract computational model used to study computability and decidability.

PRACTICE ZONE

Answer briefly.

1. Why can an abstract model be useful?
   It lets us reason about computation independently of hardware.

2. What is meant by computability?
   Whether a problem can be solved by an algorithm.

3. Give one research problem where a theoretical model would be useful.
   Proving a given language is not regular using automata theory.

I3. Advantages and Limitations

Advantages	Limitations
Clarity and precision	May ignore real-world constraints
Formal analysis	Can be highly abstract
Hardware-independent reasoning	May simplify complex systems

PART J — ALGORITHMIC APPROACHES

Designing, analysing and testing computational solutions

LEARN & REMEMBER

Algorithmic research commonly follows:

$$\text{Problem} \rightarrow \text{Algorithm} \rightarrow \text{Complexity} \rightarrow \text{Correctness} \rightarrow \text{Testing}$$

Examples include graph algorithms, sorting, searching, dynamic programming, greedy algorithms and approximation algorithms.

J1. Algorithm Research Canvas

Choose one problem.

Problem: Single-source shortest paths in a weighted graph.

Input: A weighted graph G and a source vertex s.

Output: Shortest-path distances from s to all vertices.

Algorithm idea: Dijkstra's algorithm with a min-priority queue.

Time complexity: O((V + E) log V).

Space complexity: O(V + E).

Correctness argument: Greedy choice with an inductive invariant on settled nodes.

Test cases: Positive-weight graphs; disconnected graphs.

J2. Think

Why should a researcher consider both theoretical complexity and practical execution time?

Big-O hides constants and hardware effects, so real runtime can differ; both matter.

COMMON TRAP

A successful implementation does not automatically prove optimality or correctness for every possible input.

PART K — SOFTWARE ENGINEERING APPROACHES

Applying structured development practices to research

LEARN & REMEMBER

Software engineering approaches provide systematic methods for developing and testing software systems.
Common process models include:

- Waterfall
- Agile
- Spiral

A simplified research software cycle is:

Requirements → Design → Implementation → Testing → Maintenance

K1. Research Prototype

Suppose you are developing a software prototype for a research project.

Requirements: Detect network intrusions with a low false-positive rate.

System design: Modular pipeline: capture -> features -> ML classifier.

Implementation: Python with scikit-learn and a packet parser.

Testing: Unit tests plus evaluation on a held-out dataset.

Future enhancement: Add real-time detection and a dashboard.

K2. Why Software Engineering Matters in Research

Give one reason for each:

Version control: Tracks changes and enables rollback/collaboration.

Testing: Catches bugs and confirms correct behaviour.

Documentation: Lets others understand and reuse the work.

Reproducibility: Lets others repeat and verify the results.

THINK LIKE A RESEARCHER

A research prototype may not become a commercial product. Why, nevertheless, should it be developed systematically?

So the prototype is reliable, reproducible, and easy for others to build on or verify.

PART L — MATHEMATICAL MODELLING

Representing real systems using mathematical relationships

LEARN & REMEMBER

Mathematical modelling converts a real-world system into a simplified mathematical representation. A basic process is:

System → Assumptions → Variables → Model → Analysis → Validation

Applications include network performance, reliability, queueing systems, graph problems and optimization.

L1. Build a Model

Choose one:

Network Cloud Robot AI System IoT System

System	A single-server packet-processing queue.
Research question	How does load affect average packet delay?
Assumption 1	Packet arrivals follow a Poisson process.
Assumption 2	Service times are exponential.
Variable 1	Arrival rate (lambda).
Variable 2	Service rate (mu).
Variable 3	Average number in system (L).
Relationship / equation	rho = lambda/mu; L = rho/(1 - rho).
How will you validate it?	Compare model predictions with simulation results.

L2. Simple Queueing Example

Suppose packets arrive at a server at rate λ packets/second and are processed at rate μ packets/second.

1. What does λ represent?
   The average packet arrival rate (packets/second).

2. What does μ represent?
   The average service (processing) rate (packets/second).

3. What would happen if the arrival rate becomes greater than the service rate?
   The queue grows without bound and the system becomes unstable.

PART M — PERFORMANCE ESTIMATION AND EVALUATION

Measuring efficiency, scalability and reliability

LEARN & REMEMBER

Performance evaluation assesses how effectively a system performs.
Important parameters include:

- **Time efficiency** — execution or response time
- **Space efficiency** — memory/storage
- **Scalability** — behaviour with increasing workload
- **Reliability** — consistency and failure behaviour

A basic evaluation cycle is:

Metric → Experiment → Measurement → Comparison → Interpretation

M1. Evaluation Plan

Research claim: Our model detects intrusions more accurately than the baseline.

Metric 1: Detection accuracy / F1-score.

Metric 2: False-positive rate.

Dataset / workload: CICIDS2017 network flows.

Baseline: A standard signature-based IDS.

Environment: Python on a single-GPU workstation.

Number of experiments: 10 runs with different random seeds.

Possible limitation: May not generalize to new attack types.

M2. Example

Suppose two ML models give:

Model	Accuracy	Precision	Recall	F1
A	94%	91%	88%	89%
B	92%	95%	86%	90%

1. Which model has higher accuracy? Model A (94%).
2. Which model has higher precision? Model B (95%).
3. Which model has higher F1-score? Model B (90%).
4. Can you declare one model universally "better" using only these values? Why?
   No - each wins on different metrics; the best choice depends on the goal.

PART N — PERFORMANCE TOOLS

Selecting appropriate tools for research evaluation

LEARN & REMEMBER

Examples of tools/categories used for performance studies include:

- Profilers
- Simulation tools
- Benchmarking suites
- Monitoring tools

Examples from the Unit II material include gprof, Valgrind, JProfiler, NS3, CloudSim, OMNeT++, SPEC CPU, MLPerf, Prometheus and Grafana.

PRACTICE ZONE

Match the requirement with an appropriate tool/category.

1. Measure CPU and memory behaviour of a program.
   Profiler (gprof / Valgrind / JProfiler).
2. Simulate network behaviour.
   Network simulator (NS3 / OMNeT++).
3. Compare standardized computing performance.
   Benchmark suite (SPEC CPU / MLPerf).
4. Monitor system metrics continuously.
   Monitoring tools (Prometheus / Grafana).

N1. Researcher's Performance Sheet

Experiment title:

Evaluating an ML-based network intrusion detector.

Hypothesis:

Feature selection lowers the false-positive rate without reducing recall.

Metric(s):

Precision, recall, and false-positive rate.

Observed result:

FP-rate fell from 5% to 1.8% at 95% recall.

Interpretation:

Feature selection cuts false alarms while keeping detection high.

Limitation:

Tested on one dataset; may not generalize.

PART O — INTEGRATED RESEARCH ACTIVITY

Apply multiple Unit II concepts to one research problem

CLASSROOM ACTIVITY

Scenario

A team proposes:

"Our AI-enabled smart irrigation system reduces water consumption while maintaining crop health."

You are the research-methods investigator.

Your task is to examine the claim scientifically.

O1. Dialectical Analysis

Original claim:

The AI irrigation system cuts water use while keeping crops healthy.

Counter-argument 1:

Sensor errors could mislead irrigation decisions.

Counter-argument 2:

Results may not hold across soil types or seasons.

Refined claim:

On the tested crops, the system cuts water use without harming yield.

O2. Argument Model

Which reasoning model would be appropriate?

Using observations to generalize performance — Inductive

Applying a known theorem to the system — Deductive

Finding the most likely cause of a sensor problem — Abductive

O3. Proof / Validation

Select appropriate methods:

Demonstration | (Empirical) | Mathematical

Explain your choice:

It is a performance claim, best tested by field experiments measuring water use and crop health.

O4. Algorithmic Approach

What algorithm could decide when irrigation should start?

A threshold policy (or decision tree) on soil-moisture level.

What would be its input?

Soil moisture, weather forecast, and crop type.

What would be its output?

Whether to start/stop irrigation (and for how long).

O5. Mathematical Model

Define three variables.

$x_1 =$ soil moisture level $\quad x_2 =$ water applied (litres) $\quad x_3 =$ crop water requirement

Write one assumption:

Crop water need is roughly constant within a growth stage.

Write a simple relationship/model:

Irrigate if x1 < threshold; amount = max(0, x3 - x1).

O6. Performance Evaluation

Metric 1	Water consumption (litres).
Metric 2	Crop-health index / yield.
Baseline	Fixed-schedule irrigation.
Experiment condition	Same crop and field over one season.
Expected result	Lower water use with similar crop health.

PART P — MINI RESEARCH DESIGN

Create your own small Computer Science research study

Research Title

Reducing False Positives in ML-Based Network Intrusion Detection.

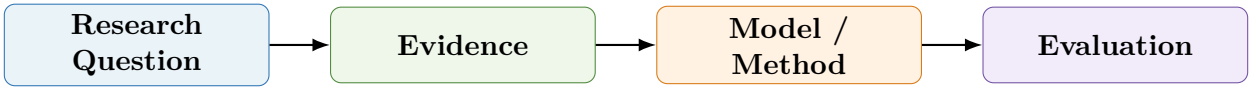
Research problem	ML-based IDS raise too many false alarms.
Research question	Can feature selection cut false positives without lowering recall?
Claim / hypothesis	Feature selection lowers the FP-rate at fixed recall.
Counter-argument	Fewer features might miss rare attacks.
Reasoning model	Inductive (empirical generalization).
Validation / proof method	Empirical evaluation on datasets.
Theoretical model	Feature space and classifier decision boundary.
Algorithmic approach	Random Forest with feature selection.
Mathematical model	FP-rate = FP / (FP + TN).
Performance metric	Precision, recall, and FP-rate.
Baseline	Full-feature classifier / signature IDS.
Expected contribution	A lighter IDS with fewer false alarms.

THINK LIKE A RESEARCHER

Researcher's check

Before accepting your own claim, ask:

1. What could prove me wrong?
2. What evidence do I need?
3. Which assumptions am I making?
4. Can another researcher reproduce my experiment?
5. Does my evidence actually support my conclusion?



UNIT II COMPLETE

Think critically. Question assumptions. Use evidence.
Build arguments. Validate claims. Evaluate results.

Research Methodology in Computer Science
School of Computer Science, UPES

RESEARCH METHODOLOGY IN COMPUTER SCIENCE

WORKSHEET – UNIT III

Literature Search, Analysis, References & Ethics

Literature Search and Academic Sources

Databases → Search Strategies → Literature Analysis → Synthesis → Mapping

Academic Integrity and Research Ethics

References → Citation → Plagiarism → Data Privacy → Ethical Research

Name: Kushagra Singh

Roll No.: 590012353

Section: B1[CCVT]

Date: 15/9/2026

School of Computer Science, UPES

Dr. Ayush Kumar Agrawal

UNIT III – WORKSHEET MAP

Literature Search, Analysis & Ethics

LEARN & REMEMBER

This unit develops the practical skills required to move from a research question to a reliable body of literature and then use that literature ethically.
The major learning areas are:

- Literature search and academic sources
- Research databases and search strategies
- Keywords and Boolean operators
- Citation chaining and search alerts
- Critical reading of research papers
- Literature analysis and synthesis
- Literature mapping and research trends
- References and citation management
- Plagiarism and effective paraphrasing
- Research ethics, data privacy and authorship
- IRB and ethical approval

No.	Learning Area	Done	Confidence
1	Literature Search	☑	5
2	Sources of Literature	☑	4
3	Literature Review	☑	5
4	Databases	☑	4
5	Search Strategies	☑	4
6	Citation Chaining	☑	5
7	Literature Analysis	☑	4
8	Literature Synthesis	☑	4
9	Literature Mapping	☑	4
10	References and Citations	☑	5
11	Plagiarism and Paraphrasing	☑	4
12	Research Ethics	☑	4
13	Data Privacy and IRB	☑	5
14	Authorship and Publication Ethics	☑	4

WARM-UP

Start Thinking Like a Researcher

PRACTICE ZONE

Think about your final-year project, research interest, or a topic you would like to investigate.

1. Where would you search for related work?
2. Would you use Google, Wikipedia, IEEE Xplore, ACM DL or something else?
3. How would you decide whether a paper is reliable?

My research topic: ML-based network intrusion detection (reducing false positives).

Where I would search first: IEEE Xplore.

Why?

It indexes peer-reviewed CS and security research with strong search filters.

1. LITERATURE SEARCH

Building the Research Foundation

1.1 What is Literature Search?

Literature search is the systematic process of finding existing research related to a research problem, question, method, technology, or domain.

It is much more than simply typing a topic into a search engine.

A good literature search helps the researcher discover:

- what is already known,
- what methods have already been used,
- which datasets and benchmarks are available,
- what results have already been achieved,
- where researchers disagree, and
- what questions remain unanswered.

LEARN & REMEMBER

Research does not start with "What can I build?"
It should first ask:

What has already been done, what is missing, and what can I contribute?

1.2 Why Literature Search Matters

Literature search:

1. helps avoid unnecessary duplication;
2. supports the novelty of a research problem;
3. identifies useful tools, datasets and benchmarks;
4. provides background for the research problem;
5. helps formulate research questions;
6. provides evidence for the proposed methodology;
7. identifies gaps and unresolved issues.

THINK LIKE A RESEARCHER

Suppose a student proposes:

"I will develop a CNN model for plant disease detection."

What should the student search before starting the work?

Existing plant-disease datasets, prior CNN methods, reported accuracies, and remaining gaps.

2. SOURCES OF LITERATURE

Academic vs. Casual Information

2.1 Primary and Secondary Sources

Source Type	Examples
Primary Sources	Original journal papers, conference papers, patents, technical reports
Secondary Sources	Review papers, survey papers, books
Digital Libraries	IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect
Indexing Databases	Scopus, Web of Science
Preprint Sources	arXiv and other research repositories
Search Engines	Google Scholar and general web search

2.2 Academic Source vs. Casual Source

Consider the following:

Casual Search	Academic Search
Blogs	Journal papers
News articles	Conference papers
YouTube videos	Technical reports
Wikipedia	Patents
Random websites	Peer-reviewed literature

COMMON TRAP

Blogs, videos and Wikipedia may be useful for learning a concept, but they should not automatically be treated as equivalent to peer-reviewed research evidence.

2.3 Mini Case Study

A student wants to research:

Blockchain for Supply Chain

The student first searches the general web and finds blogs and news articles.

After searching IEEE Xplore and other academic databases, the student finds research discussing scalability, energy consumption and security.

THINK LIKE A RESEARCHER

What changed between the two searches?

The academic search returned peer-reviewed evidence (scalability, energy, security) instead of blogs.

3. LITERATURE REVIEW

From Papers to Understanding

3.1 Objectives of a Literature Review

A literature review should not become a list of paper summaries.
It should:

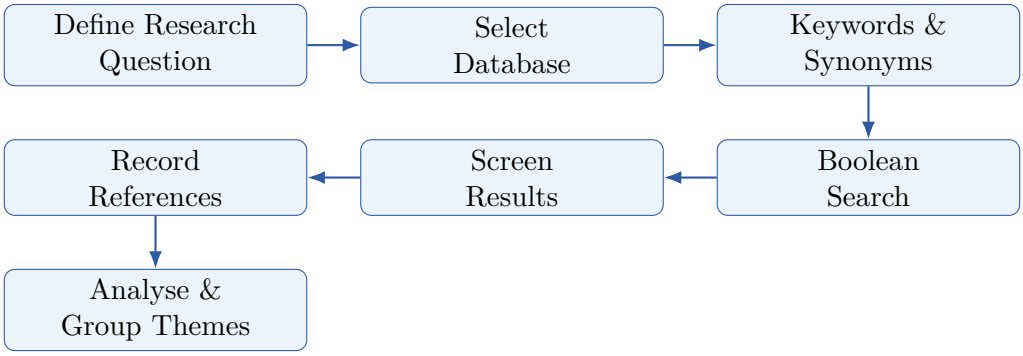
- summarize current knowledge,
- compare methods and results,
- identify research gaps,
- highlight contradictions or debates,
- establish the theoretical and methodological foundation of the study.

LEARN & REMEMBER

Weak literature review:
"Paper A did this. Paper B did this. Paper C did this."

Stronger literature review:
"Most studies use machine learning for intrusion detection; however, real-time deployment remains insufficiently explored because of computational and scalability constraints."

3.2 Literature Search Process



PRACTICE ZONE

Write the correct order for your own literature search.
Research Question: Can feature selection reduce false positives in ML-based NIDS?
Database: IEEE Xplore.
Keywords: intrusion detection, machine learning, false positives.
Search String: ("intrusion detection") AND ("machine learning") AND ("false positive")
How will I screen the papers?

By checking titles and abstracts for relevance, recency, and venue quality.

4. RESEARCH DATABASES

Finding Reliable Literature

4.1 Why Use Academic Databases?

Academic databases provide structured access to scholarly research.
They commonly support:

- author searching,
- year filtering,
- keyword searching,
- subject filtering,
- citation tracking,
- full-text access,
- export of citation information.

4.2 Major Databases in Computer Science

Database	Typical Coverage
IEEE Xplore	Computer science, electronics, AI, IoT, robotics, communications
ACM Digital Library	Computing, algorithms, software, HCI
SpringerLink	Computer science, engineering and interdisciplinary research
ScienceDirect	Broad scientific and engineering literature
Scopus	Abstracting, indexing and citation analysis
Web of Science	Indexing and citation analysis
arXiv	Preprints in AI, ML, mathematics and related areas
Google Scholar	Broad academic search with less filtering

PRACTICE ZONE

For each topic, select a suitable database.

1. Robotics and automation: IEEE Xplore.
2. Algorithms and software: ACM Digital Library.
3. AI and machine learning: arXiv / IEEE Xplore.
4. Citation analysis: Scopus / Web of Science.

5. KEYWORD SELECTION

Search with Precision

5.1 What Makes a Good Keyword?

A good keyword should be:

- technically relevant,
- specific to the research problem,
- commonly used in the field,
- supported by synonyms or alternative terminology.

5.2 Example

Research topic:

AI in Cybersecurity

Possible keywords:

Artificial Intelligence

Machine Learning

Intrusion Detection

Malware Detection

Deep Learning

LEARN & REMEMBER

A useful technique is to inspect the keywords used in good papers that you have already found. Those keywords can then be used to expand your search.

PRACTICE ZONE

Develop keywords for:

Topic: Drone Applications in Agriculture

Keyword 1	Drone / UAV
Keyword 2	Precision agriculture
Keyword 3	Crop monitoring
Synonym	Unmanned aerial vehicle
Related term	Remote sensing

5.3 Search Vocabulary

Think beyond one word.
For example:

IoT Security

may be expanded into:

IoT | Internet of Things | IoT Security | Network Security | Cybersecurity

6. BOOLEAN SEARCH

AND, OR, NOT and Advanced Search

6.1 Boolean Operators

Operator	Purpose	Example
AND	Narrows the search	AI AND Healthcare
OR	Broadens the search	IoT OR "Internet of Things"
NOT	Excludes a term	Cloud NOT Gaming
" "	Searches an exact phrase	"deep reinforcement learning"
*	Wildcard / word variation	comput*

6.2 Understanding the Difference

Suppose the researcher searches:

$$AI$$

The result may be very broad.
Now:

$$AI\ AND\ Healthcare$$

becomes more focused.
Now:

(Artificial Intelligence OR Machine Learning)AND(Healthcare OR Medical Diagnosis)

captures multiple related terms.

PRACTICE ZONE

Create a Boolean search string for:

Machine Learning for Crop Disease Detection

Your search string:

("machine learning" OR "deep learning") AND ("crop disease" OR "plant disease detection")

7. SEARCH STRATEGY

Search, Screen, Record

7.1 Screening Search Results

Do not download every result.
First examine:

1. Title
2. Abstract
3. Keywords
4. Publication year
5. Venue
6. Relevance to your research question

Only then decide whether the full paper deserves detailed reading.

COMMON TRAP

A common mistake is downloading hundreds of papers and reading none of them carefully. Quality of analysis is more important than the number of PDFs collected.

7.2 Search Record

Maintain a simple search record.

Date	Database	Search Query	Results / Notes

7.3 Common Search Mistakes

- Using only one or two keywords.
- Relying only on general web search.
- Ignoring Boolean operators.
- Not recording search strategies.
- Skipping abstracts.
- Collecting too many irrelevant papers.

8. CITATION CHAINING

Finding Connected Research

8.1 Backward Citation Chaining

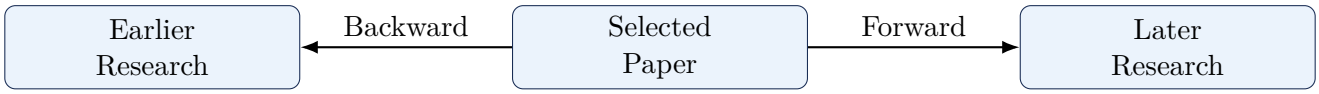
Backward chaining means examining the references of an existing paper.
It answers:

Which earlier studies influenced this paper?

8.2 Forward Citation Chaining

Forward chaining means finding newer papers that cite the selected paper.
It answers:

What research came after this work?



PRACTICE ZONE

You find one highly relevant paper.

1. Where will you look for older foundational studies?
2. Where will you look for newer developments?

Answer:

Backward chaining (references) for older studies; forward chaining (citing papers) for newer ones.

8.3 Search Alerts

Many databases allow researchers to create alerts for:

- new papers containing selected keywords,
- new publications by an author,
- new citations of a paper,
- new research in a particular topic.

THINK LIKE A RESEARCHER

If your research project lasts six months, how could search alerts help you?

They notify me of new relevant papers and citations, keeping the review current throughout the project.

9. CRITICAL READING OF A RESEARCH PAPER

Read Beyond the Abstract

9.1 Why Critical Reading?

Collecting papers is not enough.
Critical reading helps determine:

- whether the study actually addresses your problem,
- whether the methodology is appropriate,
- whether the dataset is suitable,
- whether the results support the claims,
- what limitations remain.

9.2 Paper Reading Sequence

Section	Questions to Ask
Abstract	What problem and contribution are presented?
Introduction	Why is the problem important? What research question is addressed?
Methodology	What algorithm, dataset or experimental design is used?
Results	What evidence is presented? What are the performance measures?
Discussion	What do the results actually mean?
Limitations	What did the authors not solve?
References	Which earlier studies are important?

9.3 Paper Analysis Sheet

Research Problem	High false-positive rates in network intrusion detection.
Research Question	Can ML cut false positives while keeping recall high?
Method / Algorithm	Random Forest classifier on flow features.
Dataset / Experimental Setup	CICIDS2017; 70/30 train-test split.
Main Result	High detection with a low false-positive rate.
Performance Metric	Accuracy, precision, recall, FP-rate.
Limitation	Evaluated on one dataset; offline only.
Future Scope	Real-time, on-device deployment and new attacks.

10. LITERATURE ANALYSIS

Evaluate Individual Studies

10.1 Analysis vs. Summary

A summary tells what a paper contains.
Analysis examines:

- strengths,
- weaknesses,
- methods,
- datasets,
- results,
- assumptions,
- limitations.

LEARN & REMEMBER

Summary:
"Paper A uses SVM for intrusion detection."

Analysis:
"Paper A uses SVM with the KDD Cup 99 dataset and achieves 92% accuracy. The method is computationally simple, but the dataset may not represent modern network traffic."

10.2 Comparative Analysis

Consider three studies:

	Paper A	Paper B	Paper C
Method	SVM	Deep Learning	Hybrid
Dataset	KDD Cup 99	NSL-KDD	NSL-KDD
Accuracy	92%	96%	High
Complexity	Low	Medium	High
Scalability	Good	Moderate	Limited

THINK LIKE A RESEARCHER

Which method would you investigate further if:

1. accuracy is the main concern?
2. computational cost is the main concern?
3. real-time deployment is the main concern?

Explain your reasoning.

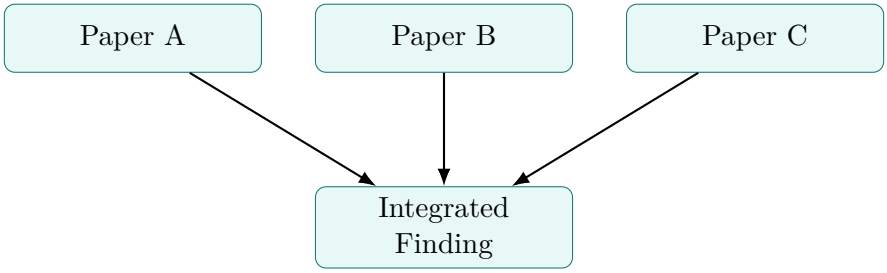
Accuracy: Paper B (96%). Cost: Paper A (low complexity). Real-time: Paper A (good scalability).

11. LITERATURE SYNTHESIS

From Individual Papers to the Field

11.1 What is Synthesis?

Analysis breaks individual studies into components.
Synthesis combines findings across multiple studies to understand the larger research landscape.



11.2 Techniques for Literature Synthesis

Technique	Purpose
Thematic Synthesis	Group studies around common themes
Methodological Comparison	Compare algorithms, datasets and experimental approaches
Chronological Review	Show how research has evolved over time
Conceptual Mapping	Connect major concepts and relationships
Systematic Review	Use a structured process to identify and synthesize literature

11.3 Finding the Research Gap

Possible research gaps may appear when:

- studies produce contradictory findings,
- datasets are outdated,
- real-world validation is missing,
- scalability is ignored,
- cost or computational limitations are ignored,
- ethical aspects are not considered,
- a method works in theory but not in practical deployment.

PRACTICE ZONE

From the three papers discussed earlier, complete:

What has been solved?

ML methods reach high accuracy for intrusion detection on benchmark datasets.

What remains unsolved?

Real-time, scalable detection on modern network traffic is still limited.

Possible research gap:

Low-latency, low-false-positive detection validated on current traffic.

12. LITERATURE MAPPING

Seeing the Big Picture

12.1 What is Literature Mapping?

Literature mapping is the process of systematically or visually organizing research papers to identify:

- themes,
- clusters,
- influential studies,
- relationships between topics,
- research trends,
- research gaps.

12.2 Why Map Literature?

When a researcher has 30, 50 or 100 papers, a simple list becomes difficult to understand.
A map helps answer:

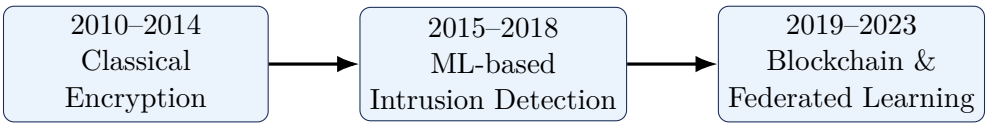
What are the major themes?
Which papers are connected?
How has the topic evolved?
Where is the research gap?

12.3 Types of Literature Mapping

Mapping Type	Description
Thematic	Group papers according to recurring themes
Chronological	Organize research according to time
Methodological	Compare methods such as ML, DL, SVM, CNN, etc.
Citation	Track citation relationships and influential papers
Conceptual	Connect concepts using diagrams or mind maps

13. LITERATURE MAPPING EXAMPLE — IoT Security

Consider the evolution of research on IoT security:



This type of mapping can reveal a shift from classical security methods towards AI-based and decentralized approaches.

13.1 Mapping Tools

Tool	Use
VOSviewer	Keyword, co-occurrence and citation maps
CiteSpace	Citation networks and research trends
Connected Papers	Exploring related research papers
XMind / MindMeister	Manual thematic mapping
Spreadsheets	Simple manual literature organization

CLASSROOM ACTIVITY

Mapping Activity
Take five papers on one topic.
For each paper, identify:

1. Main theme
2. Method
3. Year
4. Important keyword

Then group papers that have something in common.
My theme groups:

Group 1: classical encryption; Group 2: ML-based detection; Group 3: blockchain / federated learning.

14. HANDS-ON LITERATURE MAPPING

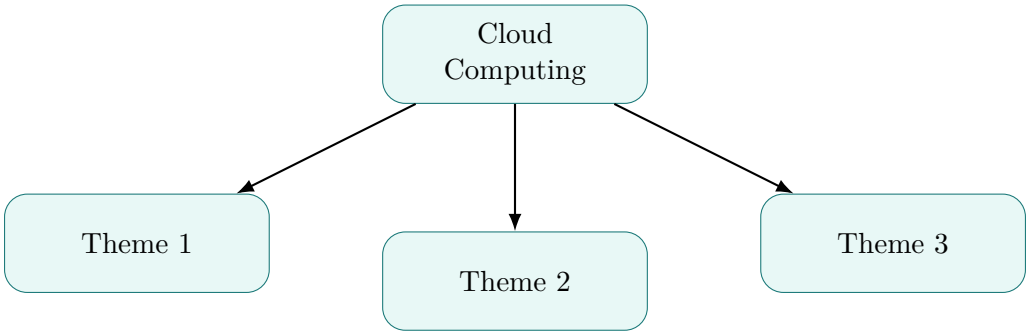
From Papers to Clusters

CLASSROOM ACTIVITY

Topic: Cloud Computing
Assume that you have five papers.
Identify three recurring themes such as:

- Cost efficiency
- Scalability
- Energy consumption

Create your own thematic map below.



Theme 1: Cost efficiency.
Theme 2: Scalability.
Theme 3: Energy consumption.

THINK LIKE A RESEARCHER

What research gap could become visible after mapping these papers?

Few studies jointly optimize cost, scalability, and energy use in cloud systems.

14.1 Best Practices for Literature Mapping

- Start mapping early.
- Combine manual notes with digital tools.
- Update the map periodically.
- Focus on influential work as well as recent work.
- Map ideas and relationships, not only keywords.

15. REFERENCES AND CITATIONS

Academic Credibility

15.1 Why References Matter

References serve several purposes:

- acknowledge original ideas,
- allow readers to verify claims,
- help avoid plagiarism,
- establish scholarly context,
- strengthen the credibility of research.

15.2 Citation Styles

Style	Typical Approach
IEEE	Numbered citations such as [1], [2]
APA	Author–year such as (Smith, 2020)
ACM	Author–year or numeric depending on venue
Chicago / Harvard	Commonly used outside core CS venues

LEARN & REMEMBER

In computer science, IEEE and ACM styles are particularly common. Always check the specific journal, conference or university guidelines before formatting references.

15.3 Citation in Writing

Examples:
Direct citation:
According to Smith [1], deep learning has applications in NLP.
Indirect citation:
Deep learning has been widely applied to natural language processing [1].
Multiple citations:
Several studies have investigated this problem [1]–[4].

COMMON TRAP

Do not mix IEEE, APA and other citation formats randomly within the same document.

16. REFERENCE MANAGEMENT

Manage Literature Systematically

16.1 Reference Management Tools

Tool	Typical Use
Mendeley	Reference management and PDF annotation
Zotero	Browser-based collection and citation management
EndNote	Academic reference management
BibTeX	Reference management for LaTeX documents
RefWorks / Citavi	Institutional reference management

16.2 Why Use Reference Managers?

Manual reference management can cause:

- missing references,
- incorrect author names,
- inconsistent formatting,
- forgotten citations,
- duplicate entries.

Reference managers help organize papers and generate citations according to a selected style.

PRACTICE ZONE

Reference Formatting Activity

Given:

Smith, J. (2021). "AI for Cybersecurity." Journal of Computer Science, 45(3), 120–135.

Write the reference in IEEE style.

[1] J. Smith, "AI for Cybersecurity," J. Comput. Sci., vol. 45, no. 3, pp. 120-135, 2021.

Now write it in APA style.

Smith, J. (2021). AI for cybersecurity. Journal of Computer Science, 45(3), 120-135.

17. CITATION ETHICS

Cite Fairly and Accurately

17.1 Good Citation Practices

A researcher should:

- cite the original source where possible,
- cite relevant work,
- include classical work when important,
- include recent work when appropriate,
- acknowledge contributions fairly,
- maintain consistent formatting.

17.2 Citation Padding

Citation padding occurs when irrelevant references are added simply to increase the number of citations.

THINK LIKE A RESEARCHER

Consider:
"Our algorithm is efficient [1–12]."
However, only two of the twelve papers actually discuss algorithmic efficiency.
Is this good citation practice?
Explain.

No; it cites irrelevant work and misrepresents the support for the claim.

17.3 Common Reference Mistakes

- Mixing citation styles.
- Missing references in the bibliography.
- Listing references that are never cited.
- Incorrect author/year/page information.
- Over-citing trivial facts.
- Citing a secondary source instead of the original source.

18. PLAGIARISM

Academic Integrity

18.1 What is Plagiarism?

Plagiarism means using another person's words, ideas or work without appropriate acknowledgment. It can involve:

- written text,
- figures,
- diagrams,
- computer code,
- datasets,
- ideas,
- previously published work.

Plagiarism may be intentional or accidental.

18.2 Types of Plagiarism

Type	Description
Direct / Verbatim	Copying text word-for-word
Mosaic	Combining copied phrases with your own wording
Self-Plagiarism	Reusing previous work without proper acknowledgment
Paraphrasing Plagiarism	Changing words while retaining the original structure
Accidental Plagiarism	Incorrect or missing citation

18.3 Plagiarism in Computer Science

Examples include:

- copying code from GitHub without credit,
- using another researcher's figure without citation,
- submitting the same project for two courses,
- reusing a previous publication as a new work without proper disclosure.

COMMON TRAP

Plagiarism is not limited to copied paragraphs. Figures, code, data and ideas may also require acknowledgment.

19. PLAGIARISM DETECTION

Similarity is a Warning Signal

19.1 Common Detection Tools

Examples include:

- Turnitin
- iThenticate
- PlagScan
- Grammarly similarity features
- MOSS for source-code similarity
- JPlag for source-code comparison

LEARN & REMEMBER

A similarity report is a tool for investigation. A similarity percentage should not be interpreted without examining what material has been matched and why.

19.2 Identify the Problem

Classify each situation:

1. A student copies a paragraph without citation.
2. A student uses a figure from a paper and cites the paper.
3. A student copies GitHub code and gives no attribution.
4. A student paraphrases an idea and cites the original source.
5. A student submits an old report as a new assignment without disclosure.

No.	Ethical / Unethical?	Reason
1	Unethical	Plagiarism (no citation).
2	Ethical	Properly cited figure use.
3	Unethical	Code plagiarism (no attribution).
4	Ethical	Correct paraphrase with citation.
5	Unethical	Self-plagiarism (no disclosure).

20. PARAPHRASING

Reword + Restructure + Cite

20.1 What is Paraphrasing?

Paraphrasing means expressing another author's idea in your own words and sentence structure while preserving the original meaning.

Good paraphrasing requires:

1. understanding the source,
2. writing the idea independently,
3. changing the structure,
4. preserving the meaning,
5. citing the source.

20.2 Example

Original:

> "Machine learning models require large labeled datasets for training."

Poor paraphrasing:

> "Machine learning systems need big labeled datasets for training."

The sentence structure remains almost the same.

Better paraphrasing:

> Effective machine-learning approaches often depend on extensive annotated datasets during model development [1].

COMMON TRAP

Simply replacing words with synonyms is not sufficient. The idea must be understood and expressed independently.

20.3 Practice

Original:

> "Cloud computing improves scalability but raises security concerns."

Write your own version.

Although cloud computing improves scalability, it also introduces new security risks [1].

Citation marker: [1]

21. ACADEMIC WRITING ETHICS

Originality and Honesty

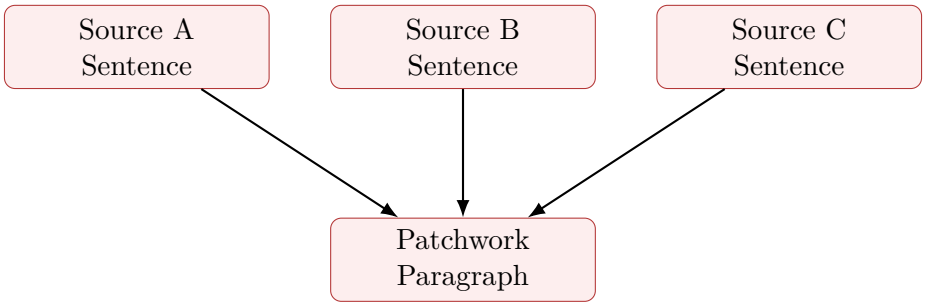
21.1 Ethical Academic Writing

A researcher should:

- cite original sources,
- use quotation marks for direct quotations,
- paraphrase rather than patch-copy,
- acknowledge collaborators,
- acknowledge prior work,
- maintain originality.

21.2 Patchwork Writing

Patchwork writing occurs when a researcher takes sentences or phrases from multiple sources and joins them together with minor changes.



THINK LIKE A RESEARCHER

Why is adding citations to every sentence not automatically enough to make patchwork writing acceptable?

Because the wording and structure are still copied; real writing expresses ideas in your own words.

22. RESEARCH ETHICS

Responsible Conduct of Research

22.1 What is Research Ethics?

Research ethics refers to principles that guide responsible conduct during research.
It promotes:

- honesty,
- transparency,
- fairness,
- responsibility,
- protection of participants,
- reliability of the scientific record.

In computer science, ethics can involve:

- software,
- algorithms,
- datasets,
- artificial intelligence,
- human-computer interaction,
- user studies.

22.2 Core Ethical Principles

Principle	Meaning
Honesty	No fabrication, falsification or misrepresentation
Objectivity	Avoid inappropriate bias in research design and interpretation
Integrity	Follow professional and institutional guidelines
Respect	Protect participants and their information
Accountability	Take responsibility for research work and contributions

23. ETHICAL ISSUES IN CS RESEARCH

People, Data and Algorithms

23.1 Major Ethical Areas

Area	Concern
Human Subjects	Consent, privacy and participant safety
Data Handling	Security, anonymization and lawful use
Authorship	Fair credit for genuine contributions
Publication	No duplicate publication or fake peer review
AI Bias	Avoid unjustified discrimination or harmful bias

23.2 Fabrication and Falsification

Fabrication means creating data or results that never existed.

Falsification means manipulating research materials, data, methods or results so that the research record is misrepresented.

COMMON TRAP

Changing experimental results simply to make an algorithm appear better is a serious ethical violation.

PRACTICE ZONE

A researcher obtains 82% accuracy but reports 94% accuracy in the paper because the higher value appears more impressive.
Identify the ethical violation and explain why it is harmful.

Falsification of results; it misleads readers, wastes others' effort, and corrupts the scientific record.

24. ETHICS IN DATA USAGE

Privacy and Responsible Data

24.1 Responsible Data Practices

When human or sensitive data is involved, researchers should consider:

- informed consent,
- privacy,
- secure storage,
- anonymization,
- pseudonymization where appropriate,
- lawful data collection,
- transparency about data usage.

24.2 Data Privacy Principles

The source material highlights frameworks such as:

- GDPR
- HIPAA
- Indian IT Act
- Digital Personal Data Protection Act (DPDP Act)

LEARN & REMEMBER

A simple principle to remember:

Collect only what is necessary, protect what you collect, and clearly explain how it will be used.

24.3 Privacy Activity

You are developing a mobile application to track student study habits.
Possible data:

- Name
- Age
- Study duration
- Location
- Device information
- Academic performance

PRACTICE ZONE

Which data is actually necessary?

Only study duration and academic performance; drop name, location, and device info.

How would you anonymize the data?

Replace names with random IDs and remove location and device identifiers.

How would you obtain informed consent?

Explain the purpose and data use, then get opt-in agreement before collecting data.

25. AUTHORSHIP ETHICS

Who Should Be an Author?

25.1 Responsible Authorship

Authorship should reflect meaningful contributions to the research.
Examples of significant contributions may include:

- research design,
- substantial experimentation,
- methodology development,
- data analysis,
- substantial interpretation,
- significant manuscript contribution.

25.2 Unethical Authorship

Gift authorship:
Adding a person as an author despite little or no contribution.
Ghost authorship:
A significant contributor is not properly identified as an author.

COMMON TRAP

Being a supervisor, senior person or friend does not automatically justify authorship. Authorship should follow the contribution criteria and the relevant journal/conference policy.

PRACTICE ZONE

Four students work on a project:

A	Research design and data analysis
B	Software implementation
C	Presentation design only
D	Data collection and manuscript preparation

Discuss which contributions could justify authorship.

A, B, and D justify authorship (design/analysis, implementation, data & writing); C (slides only) does not.

26. PUBLICATION ETHICS

Responsible Scholarly Communication

26.1 Common Ethical Violations

Examples include:

- plagiarism,
- self-plagiarism,
- fabrication,
- falsification,
- duplicate publication,
- unjustified authorship,
- fake peer review,
- hiding important conflicts or limitations.

26.2 Ethical Decision Activity

For each situation, identify the main issue.

	Situation	Issue
1	The same manuscript is submitted simultaneously to two conferences.	Duplicate submission.
2	A friend is added as an author without contribution.	Gift authorship.
3	A negative experimental result is hidden.	Selective reporting / falsification.
4	A dataset is used without checking its provenance.	Improper data provenance.
5	A collaborator's major contribution is not acknowledged.	Ghost authorship.

27. RESEARCH ETHICS CASE STUDIES

Learning from Real Situations

27.1 Case Study – Facebook Emotion Study

The source material discusses a large-scale Facebook experiment in which users' news feeds were modified to study emotional effects.

The case raises questions about:

- informed consent,
- participant autonomy,
- privacy,
- ethical approval,
- large-scale experimentation.

THINK LIKE A RESEARCHER

Discuss

If an experiment can affect a person's behaviour or emotional state, what responsibilities should researchers have before conducting it?

Get informed consent and ethics/IRB approval, minimize harm, allow withdrawal, and be transparent.

27.2 Case Study – Data Privacy

The source material discusses cases where large-scale user data created privacy risks.
A key lesson is:

Publicly accessible or apparently anonymized data can still create privacy risks.

PRACTICE ZONE

A researcher downloads a public dataset containing detailed user activity.
Before using it, what should the researcher check?

1. Source and provenance
2. Presence of personal identifiers
3. Terms of use
4. Privacy risks
5. Permission/consent where relevant

Additional consideration:

Check whether individuals can be re-identified even if the data looks anonymized.

28. IRB AND ETHICAL APPROVAL

Protecting Human Participants

28.1 What is IRB?

IRB stands for:

Institutional Review Board

It may also be referred to as an ethics committee.
Its role includes reviewing research involving human participants to ensure that:

- participants are treated appropriately,
- risks are considered,
- consent is handled properly,
- privacy is protected,
- ethical requirements are satisfied.

28.2 When May Approval Be Needed?

Examples include:

- surveys involving people,
- interviews,
- experiments involving participants,
- HCI studies,
- software usability studies,
- studies involving sensitive datasets,
- health, financial or biometric data,
- identifiable personal information.

COMMON TRAP

Do not assume that a project is "only computer science" and therefore automatically exempt from ethical review. Human participants and identifiable or sensitive data can introduce ethical requirements.

29. PRINCIPLES OF ETHICAL APPROVAL

IRB and Participants

29.1 Informed Consent

Participants should understand relevant information about the study, including its purpose, procedures and relevant risks or benefits.

29.2 Confidentiality

Personal information should be protected and not unnecessarily disclosed.

29.3 Minimal Risk

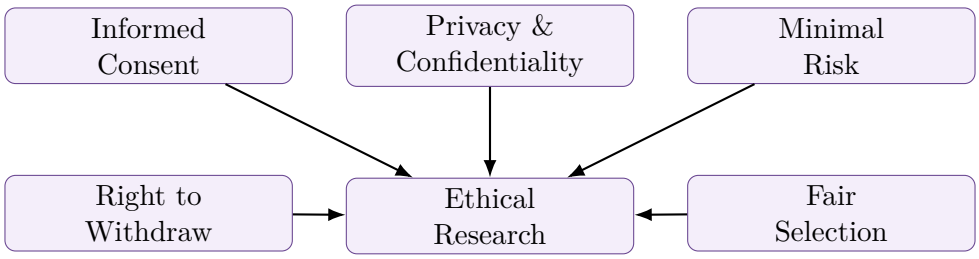
Researchers should minimize avoidable harm, discomfort or risk.

29.4 Right to Withdraw

Where applicable, participants should be able to leave a study without inappropriate consequences.

29.5 Fair Selection

Participant recruitment should be fair and should not create unjustified discrimination or exploitation.



30. INTEGRATED RESEARCH ACTIVITY

From Search to Ethical Research

CLASSROOM ACTIVITY

Choose one research topic from your discipline.
You will now move through the complete Unit III workflow.

Step 1 – Define the Research Problem

Topic:

Reducing false positives in ML-based network intrusion detection.

Research Question:

Can feature selection lower the false-positive rate without reducing recall?

Step 2 – Develop Keywords

Primary keyword	intrusion detection
Synonym	network anomaly detection
Related keyword	false-positive reduction
Technical term	machine-learning classifier

Step 3 – Create Search String

("intrusion detection") AND ("machine learning") AND ("false positive" OR "feature selection")

Step 4 – Select One Important Paper

Title	Feature Selection for ML-Based Network Intrusion Detection
Author(s)	A. Researcher et al.
Year	2021
Database	IEEE Xplore
Method	Random Forest with feature selection
Main Finding	Fewer features cut false positives while keeping recall
Limitation	Evaluated on a single dataset

31. INTEGRATED PAPER ANALYSIS

Analyse, Compare and Identify

Step 5 – Compare Two Papers

	Paper 1	Paper 2
Research Problem	High false positives in IDS	Detection accuracy in IDS
Method	SVM	Deep learning
Dataset	KDD Cup 99	NSL-KDD
Main Result	92% accuracy	96% accuracy
Strength	Simple and fast	High accuracy
Limitation	Outdated dataset	Higher compute cost

Step 6 – Synthesis

Complete:

Across the two papers, the common finding is:

Both apply ML to network intrusion detection on benchmark datasets.

The major difference is:

SVM is simpler; deep learning is more accurate but costlier.

A possible research gap is:

Low-cost methods that still reach high accuracy in real time.

Step 7 – Citation Chaining

One earlier paper I would investigate:

A foundational KDD Cup 99 intrusion-detection study.

One newer paper that cites the selected work:

A recent deep-learning IDS paper citing this work.

32. ETHICAL RESEARCH DESIGN

Apply Research Ethics

Continue with the same research topic.

32.1 Does Your Research Involve People?

Human participants involved?	☐ Yes	☑ No
Identifiable data involved?	☑ Yes	☐ No
Sensitive data involved?	☐ Yes	☑ No
Survey/interview involved?	☐ Yes	☑ No
User study involved?	☐ Yes	☑ No

32.2 Ethical Safeguards

If participants or sensitive data are involved, specify:

Consent procedure:

Use publicly released datasets; no direct human consent needed.

Data protection method:

Store data on encrypted, access-controlled machines.

Anonymization method:

Mask or hash IP addresses before analysis.

Who will have access to the data?

Only the research team.

Potential ethical risk:

Possible re-identification from network metadata.

32.3 Authorship Plan

Who may contribute significantly to the research?

Research Design	Lead student researcher.
Implementation	Student developer.
Data Collection	Student.
Data Analysis	Student.
Writing	Student and co-author.
Supervision / Guidance	Faculty supervisor.

33. UNIT III RESEARCH CHALLENGE

Search → Analyse → Map → Research Ethically

CLASSROOM ACTIVITY

Scenario

You are planning a research project on:

"AI-Based Smart Healthcare Monitoring"

Prepare a mini research-search plan.

A. Literature Search

Three databases I would search:

1. IEEE Xplore
2. PubMed / ScienceDirect
3. ACM Digital Library

Four keywords:

1. smart healthcare
2. remote patient monitoring
3. machine learning
4. wearable sensors

Boolean search string:

("smart healthcare" OR "remote monitoring") AND ("machine learning" OR "deep learning")

B. Literature Analysis

What methods would you compare?

CNN, LSTM, and classical ML for vital-sign prediction.

What datasets would you investigate?

MIMIC-III and wearable-sensor datasets.

What limitations would you look for?

Small samples, privacy limits, and poor generalization.

C. Literature Mapping

Which mapping would be useful?

☐ Thematic ☐ Chronological ☑ Methodological

☐ Citation ☐ Conceptual

Why?

It compares ML methods across studies, which fits a method-focused review.

34. ETHICS APPLICATION

Complete the Researcher Mindset

Continue with the healthcare research scenario.

34.1 Ethical Risks

Identify at least four:

1. Breach of patient privacy.
2. Data misuse or re-identification.
3. Biased models across patient groups.
4. Inadequate informed consent.

34.2 Data Protection

What personal information should be avoided if it is not necessary?

Names, exact addresses, and other direct identifiers.

How will sensitive information be protected?

Encryption, access control, and anonymization.

34.3 Participant Consent

Write a short statement explaining what a participant should know before joining the study.

The study's purpose, what data is collected, how it is used and protected, and their right to withdraw.

34.4 Research Integrity

Complete the statements:

1. I will not fabricate data or results.

1. I will not falsify methods or findings.

1. I will acknowledge all sources and contributors.

1. I will protect participants' data and privacy.

1. I will report limitations honestly and in full.

LEARN & REMEMBER

A strong researcher searches systematically, analyses critically, maps knowledge, cites honestly, protects data and conducts research responsibly. **A strong researcher does not only find information. A strong researcher searches systematically, analyses critically, maps knowledge, cites honestly, protects data and conducts research responsibly.**

LITERATURE SURVEY TABLE

Research Methodology in Computer Science

Name: Kushagra Singh **Roll No.:** 590012353

Section: B1[CCVT] **Date:** 15/9/2026

Assignment Task
Select one research topic and find **7 relevant research papers**. Read the papers and prepare a Literature Survey Table.
Use **short points or phrases** in the table and write the information in your **own words**.

How to Fill the Literature Survey Table

1. **Search:** Find relevant papers using IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, Scopus or Google Scholar.
2. **Citation:** Write the author(s) and publication year.
3. **Objective / Problem:** Write what problem the researchers are trying to solve or study.
4. **Methodology:** Write the method, algorithm, model, experiment, dataset or technique used.
5. **Key Finding:** Write the main result or contribution of the paper.
6. **Limitation:** Write an important limitation, unanswered question or future research area mentioned in the paper.

Research Topic: Machine Learning for Network Intrusion Detection and False-Positive Reduction

References / Citations

Write the complete reference of the seven papers used above. Include author(s), year, paper title, journal/conference, and DOI or link where available.

1. [1] A. L. Buczak and E. Guven, "A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection," IEEE Commun. Surveys Tuts., vol. 18, no. 2, pp. 1153-1176, 2016, doi: 10.1109/COMST.2015.2494502.
2. [2] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A Detailed Analysis of the KDD CUP 99 Data Set," in Proc. IEEE CISDA, 2009, pp. 1-6, doi: 10.1109/CISDA.2009.5356528.
3. [3] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in Proc. ICISSP, 2018, pp. 108-116, doi: 10.5220/0006639801080116.
4. [4] R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep Learning Approach for Intelligent Intrusion Detection System," IEEE Access, vol. 7, pp. 41525-41550, 2019, doi: 10.1109/ACCESS.2019.2895334.
5. [5] C. Yin, Y. Zhu, J. Fei, and X. He, "A Deep Learning Approach for Intrusion Detection Using Recurrent Neural Networks," IEEE Access, vol. 5, pp. 21954-21961, 2017, doi: 10.1109/ACCESS.2017.2762418.
6. [6] B. A. Tama, M. Comuzzi, and K.-H. Rhee, "TSE-IDS: A Two-Stage Classifier Ensemble for Intelligent Anomaly-Based Intrusion Detection System," IEEE Access, vol. 7, pp. 94497-94507, 2019, doi: 10.1109/ACCESS.2019.2928048.
7. [7] M. A. Ambusaidi, X. He, P. Nanda, and Z. Tan, "Building an Intrusion Detection System Using a Filter-Based Feature Selection Algorithm," IEEE Trans. Comput., vol. 65, no. 10, pp. 2986-2998, 2016, doi: 10.1109/TC.2016.2519914.

No.	Citation	Objective / Problem	Methodology	Key Finding	Limitation
1	Buczak & Guven (2016)	Survey ML/data-mining methods for cyber intrusion detection.	Review of ML techniques (SVM, trees, clustering); compares studies and datasets.	Maps method strengths and gaps; stresses labeled-data and real-time needs.	Survey only; notes lack of current datasets and high false-alarm rates.
2	Tavallaee et al. (2009)	Expose flaws in the KDD'99 IDS benchmark.	Statistical analysis of KDD'99; propose refined NSL-KDD; test classifiers.	Redundant records bias results; NSL-KDD gives fairer evaluation.	Still synthetic, dated traffic; not fully like modern networks.
3	Sharafaldin et al. (2018)	Create a realistic, labeled modern IDS dataset.	Built CICIDS2017 with real attacks; extracted flow features; tested ML models.	Provides up-to-date benchmark; finds informative flow features.	Class imbalance; single testbed may limit generalization.
4	Vinayakumar et al. (2019)	Scalable deep-learning IDS for host and network.	Deep neural nets vs classical ML on KDD, NSL-KDD, UNSW, CICIDS.	DNNs generalize across datasets with competitive detection.	High compute/training cost; tuning needed; false alarms remain.
5	Yin et al. (2017)	Improve IDS accuracy with deep sequence models.	Recurrent neural network (RNN-IDS) on NSL-KDD; binary and multiclass.	RNN beats traditional ML in detection accuracy.	Longer training; evaluated mainly on NSL-KDD.
6	Tama et al. (2019)	Boost anomaly detection and cut false alarms.	Two-stage classifier ensemble + hybrid feature selection; NSL-KDD & UNSW-NB15.	Higher detection and accuracy than single models.	Added model complexity; only two datasets tested.
7	Ambusaidi et al. (2016)	Select informative features to improve IDS efficiency.	Filter-based mutual-information feature selection with LS-SVM classifier.	Fewer features give faster, accurate detection.	Older datasets (KDD'99/NSL-KDD); filter ignores feature interaction.

Before Submission:

☑ 7 relevant papers selected ☑ Citations recorded

☑ Table completed ☑ Information written in own words ☑ References / links provided

Search → Read → Cite → Fill the Table

Research Methodology in CS - Mid-Semester Examination

Answers Only

Q1. Steps involved in a research process

1. Identify and define the research problem clearly.
2. Review existing literature to understand prior work and identify gaps.
3. Formulate research questions, objectives and, where appropriate, hypotheses.
4. Design the methodology: research approach, variables, sampling, tools and procedures.
5. Collect relevant and reliable data systematically.
6. Analyze the data using suitable qualitative, quantitative or computational methods.
7. Interpret the results in relation to the objectives and existing knowledge.
8. Draw conclusions, state limitations and suggest future work.
9. Report and communicate the research with proper citations and ethical practices.

Q2. Necessity of defining a research problem

A clearly defined research problem gives the study direction and prevents vague or unfocused investigation. It specifies what is to be studied, helps establish scope and objectives, guides the choice of methodology and data, and makes it possible to judge feasibility. A precise problem also supports literature searching, hypothesis formulation and meaningful interpretation of results. Without proper definition, time and resources may be spent collecting irrelevant evidence or answering the wrong question.

Q3. Plagiarism and how it can be avoided

Plagiarism: presenting another person's words, ideas, code, figures, data or other intellectual work as one's own without appropriate acknowledgement.

Avoidance: cite every borrowed idea or result; use quotation marks and citations for exact wording; paraphrase genuinely by understanding and restructuring the idea rather than merely replacing words; cite the original source even after paraphrasing; keep accurate reference records; acknowledge reused code, figures and datasets; and check the final work for missing or inconsistent citations.

Q4. Fact-finding, analysis and evaluation

Yes. Research depends on more than collecting facts. Fact-finding supplies dependable evidence; analysis organizes and examines that evidence to reveal patterns, differences and relationships; and evaluation judges the significance, validity and implications of the findings. For example, in an algorithm study, merely recording execution times is fact-finding. Comparing the measurements under controlled workloads is analysis, while deciding whether the algorithm satisfies the research objective and discussing limitations is evaluation. Together these stages turn observations into defensible knowledge.

Q5. Literature review and an effective literature-study approach

Literature review: a systematic and critical examination of existing scholarly work related to a research problem. It summarizes current knowledge, compares methods and findings, identifies disagreements and limitations, and reveals research gaps.

Effective approach: define the research question; choose academic databases such as IEEE Xplore, ACM DL, Scopus or Google Scholar; develop keywords, synonyms and Boolean search strings; screen titles and abstracts using relevance criteria; read selected papers critically; record citations and key details; use backward/forward citation chaining; group studies by themes or methods; compare and synthesize findings; identify the gap; and maintain references consistently.

Q6. Important research designs used in experimental hypothesis-testing research

Important experimental designs include:
1. Completely Randomized Design (CRD): experimental units are assigned randomly to treatments. It is simple and suitable when units are reasonably homogeneous.
2. Randomized Block Design (RBD): units are divided into relatively homogeneous blocks and treatments are randomized within each block. Blocking reduces variation due to a known nuisance factor.
3. Latin Square Design: treatments are arranged so each treatment occurs once in every row and column, controlling two nuisance variables simultaneously.
4. Factorial Design: two or more factors are varied together. It estimates both individual factor effects and interactions,

making it efficient for studying complex systems.
5. Pre-test/Post-test Control Group Design: experimental and control groups are measured before and after treatment, helping estimate the treatment effect while accounting for initial differences.
6. Post-test-only Control Group Design: randomized groups are compared after treatment when a pre-test is unnecessary or may influence behavior.

A sound experimental design uses control, randomization and replication to reduce bias, estimate experimental error and permit valid testing of hypotheses.

Q7(a). Formulation and purpose of a research hypothesis

A research hypothesis is a clear, testable statement predicting a relationship or difference between variables. Formulation normally begins with the research problem and literature review, identifies relevant variables and the population/context, states the expected relationship, and converts it into testable form. Statistical studies commonly specify a null hypothesis (H0: no effect/relationship) and an alternative hypothesis (H1: an effect/relationship exists). A good hypothesis is specific, measurable, logically derived, feasible and falsifiable.

Its purpose is to focus the investigation, identify what evidence must be collected, guide research design and statistical testing, connect theory with observation, and provide an objective basis for accepting or rejecting a proposed explanation.

Q7(b). Deductive and inductive research

Deductive research moves from general theory or rule to a specific testable prediction. The researcher begins with theory, derives a hypothesis, gathers evidence and tests whether the prediction follows. If premises are true and reasoning is valid, a deductive conclusion is logically certain.

Inductive research moves from specific observations toward patterns and broader generalizations or theory. The conclusion is supported by evidence but remains probable rather than logically guaranteed.

Example: applying a proven algorithmic theorem to a particular graph is deductive; observing the same performance trend across many datasets and proposing a general rule is inductive.

Q8. Fundamental, Basic, Applied and Industrial Research

(a) Fundamental Research: seeks new principles, theories or knowledge without requiring an immediate practical application. Example: studying the theoretical limits of computation.

(b) Basic Research: expands understanding of underlying phenomena and concepts; the term is often used closely with fundamental research. Example: investigating why a new learning architecture generalizes under particular assumptions.

(c) Applied Research: uses knowledge to solve a specific practical problem. Example: developing an ML intrusion-detection model to reduce false alarms in enterprise networks.

(d) Industrial Research: research directed toward practical innovation, processes or products in an industrial setting, often under cost, reliability and deployment constraints. Example: evaluating a low-latency anomaly-detection system for a telecom operator's production network.

Q9. Inference on effectiveness of the anthrax vaccine

Observed table: inoculated goats - 15 survived, 2 died (17 total); non-inoculated goats - 4 survived, 10 died (14 total).

Survival rate among inoculated goats = 15/17 = **88.24%**.
Survival rate among non-inoculated goats = 4/14 = **28.57%**.
Death rate among inoculated goats = 2/17 = **11.76%**.
Death rate among non-inoculated goats = 10/14 = **71.43%**.

For a 2 x 2 chi-square test, the expected counts under no association are approximately 10.42, 6.58, 8.58 and 5.42. The Pearson chi-square statistic is about **11.48** with 1 degree of freedom ($p < 0.001$). Thus survival is strongly associated with inoculation in this sample. The inoculated group has a much higher survival proportion, providing strong evidence that the vaccine was effective under the conditions of the experiment.

Q9 OR (a). Necessity of defining a Research Problem

Defining the research problem converts a broad topic into a clear, researchable question. It fixes the scope, identifies relevant variables and evidence, guides objectives and methodology, helps assess novelty and feasibility, and prevents wasteful collection of irrelevant data. A well-defined problem also provides a standard against which the final results can be interpreted.

Q9 OR (b). Ways of conducting literature surveys in modern times

Modern literature surveys use academic digital libraries and indexing databases such as IEEE Xplore, ACM Digital Library, Scopus, Web of Science and Google Scholar. Researchers construct keyword/synonym lists and Boolean queries, apply year/subject filters, screen titles and abstracts, and maintain search records. Backward citation chaining finds foundational work, while forward citation chaining finds later developments. Search alerts keep the review current. Reference managers such as Zotero or Mendeley organize papers and citations, while literature-mapping tools can reveal themes, citation networks and research trends.

Q10(a). Experience survey

An experience survey is an exploratory method in which knowledgeable people are consulted to gain insight into a research problem. Experts, practitioners or experienced users may be interviewed using flexible questions. It helps clarify the problem, discover important variables, identify practical constraints and generate hypotheses when formal evidence is initially limited.

Q10(b). Pilot survey

A pilot survey is a small-scale trial conducted before the main survey or study. It tests the questionnaire, sampling procedure, wording, timing and data-collection process. It helps identify ambiguity, missing response options, technical problems and feasibility issues, allowing the researcher to revise the instrument before committing full resources.

Q10(c). Components of a research problem

A research problem normally contains: the problem or difficulty requiring investigation; the population, system or context in which it occurs; relevant variables or concepts and their possible relationships; the research gap or reason existing knowledge is insufficient; clear objectives/questions; and boundaries such as dataset, location, technology and timeframe. It should also be researchable, feasible, relevant and capable of producing analyzable evidence.

Q10(d). Rephrasing the research problem

Rephrasing means converting a broad, ambiguous or impractical problem into a precise research statement or question. The researcher clarifies key terms, narrows the population/context and variables, specifies measurable outcomes, removes unsupported assumptions and checks feasibility against literature and available resources. Example: instead of 'Improve cybersecurity,' use 'How can feature selection reduce false-positive rates of ML-based network intrusion detection on a defined benchmark dataset?'

Q11(a). Three copied sentences with changed words plus citation

Yes, it can still constitute plagiarism - specifically inadequate paraphrasing or mosaic/patchwriting - if the student preserves the source's sentence structure and merely substitutes a few words. A citation acknowledges the source of the idea but does not make near-verbatim copying acceptable. The student should understand the passage, express the idea independently with substantially different wording and structure, and retain the citation. Exact borrowed wording should be placed in quotation marks.

Q11(b). Paraphrasing, similarity scores and academic integrity

Good paraphrasing can reduce text similarity because the writer expresses the source's meaning using genuinely independent wording and sentence structure. However, reducing a similarity percentage is not the academic purpose of paraphrasing. Integrity is maintained only when the original meaning is represented accurately, the source is cited, and the writer does not disguise copied language. A low similarity score does not by itself prove originality.

Q11(c). AI tools used to paraphrase entire sections

Methodological concerns include loss or distortion of technical meaning, introduction of unsupported claims, inconsistent terminology, removal or misplacement of citations, and difficulty determining what intellectual work was performed by the researcher. Extensive AI rewriting may also obscure source dependence rather than produce genuine synthesis. The researcher must verify every rewritten claim against the source, preserve citations, comply with institutional or publisher rules on AI use, disclose use where required, and remain responsible for the final text.

Q11(d). Role of citation in paraphrasing

Citation identifies the original source of an idea, lets readers verify it, gives appropriate credit and distinguishes prior knowledge from the researcher's own contribution. Paraphrasing alone does **not** eliminate the need for citation: changing wording does not transfer ownership of the underlying idea. Unless the information is common knowledge, a faithful paraphrase still requires a citation.

Q11 OR (a). Interpreting an 18% similarity report

An 18% similarity score is neither automatically acceptable nor automatically plagiarism. It may be acceptable when matches consist mainly of properly quoted/cited text, references, standard terminology, unavoidable methodological phrases or small independent fragments. It is problematic when the matched material contains uncited copying, patchwriting, copied structure, reused work without disclosure, or substantial similarity to a particular source. Therefore the matched passages and their context must be examined rather than judging only the percentage.

Q11 OR (b). Mosaic plagiarism and similarity checks

Mosaic plagiarism, or patchwriting, combines phrases, sentence structures or ideas from one or more sources with minor alterations while presenting the result as original writing. It can escape simple similarity checks when words are substituted, sentence order is changed, material is drawn from many sources, or the tool's database does not contain the original source. Human review of wording, structure, citations and source dependence is therefore necessary.

Q11 OR (c). Improper citation despite paraphrasing

A paraphrase can still be academically improper if it cites the wrong source, cites a secondary source as though it were the original, omits the citation at the point where the borrowed idea appears, misrepresents what the cited author actually concluded, or uses a citation that does not support the claim. Proper citation must be accurate, relevant and traceable to the actual source of the idea.

Q11 OR (d). Role and limitations of similarity-detection tools

Tools such as Turnitin and iThenticate compare submitted text with material in their databases and highlight matching passages. They are useful for locating possible unattributed copying and helping reviewers investigate sources. Their limitations are important: similarity is not identical to plagiarism; properly quoted material and references may match; paraphrased or mosaic plagiarism may evade detection; database coverage is incomplete; and a percentage cannot determine intent or academic context. Final judgment therefore requires human examination of the matched passages and citations.

5900 12353
KUSHAGRA SINGH
B-1 [CCVT]

__/__/____

Research Methodology – Worksheet - 1 [Summary]

[Page - 1]

1.) **Research:**

Research is a systematic and scientific process of finding information, solving problems, and discovering knowledge using evidence.

Research Process:

Problem Identification → Data Collection → Analysis → Interpretation → Conclusion.

Objectives:

Find answers, explain relationships, test theories, and solve real – world problems.

2.) **Types of Research**

Type	Meaning
Exploratory	Discover new ideas and patterns.
Descriptive	Describe facts and characteristics
Analytical	Analyze data and relationships.
Theoretical	Models, logic, and proofs.
Empirical	Observation and experiments.
Fundamental	Develop basic knowledge.
Applied	Solve practical problems.

__/__/__

Theoretical : Prove an algorithm mathematically.

Empirical : Test a model using a dataset

3.) Research Problem:
A research problem is a specific question or issue that requires investigation.

Sources:
- Literature gaps and research papers.
- Real-world and industry problems.
- Emerging technologies.
- Previous projects and academic work.
- Research databases and patents.

Identification Steps: Explore → Narrow → Formulate → Validate.

4.) Good Research Problem:
A good research problem should be:
- Clear and concise.
- Researchable and feasible.
- Relevant and original.
- Suitable for data collection and analysis.

Common Errors:
- Too broad or too narrow
- Unclear problem.
- Lack of resources or time
- Ignoring previous research
- Lack of relevance.

5. Scope and Boundaries

Scope defines the limits of research.

Main Boundaries:

↳ Timeframe
↳ Geographic location
↳ Dataset or target population.
↳ Tools and technologies.
↳ Out-of-scope areas.

Example: Instead of "All in Smart cities", study "Traffic prediction using ML on a selected dataset."

× × ×

[Page-2]

6.) Research Objectives

Objectives explain ~~the~~ what the research aims to achieve. They provide direction and help define the methodology.

SMART Objectives:

Letter	Meaning
S	Specific — clear
M	Measurable — Can be evaluated
A	Achievable — Realistic
R	Relevant — Related to problem
T	Time-bound — Fixed time

Example: Measure the average latency of an algorithm on a dataset within four weeks.

7.) Data Types

Based on Source:

↳ Primary Data → Collected directly (interviews, sensor readings, experiments).

↳ Secondary Data → Existing information (Kaggle datasets, IEEE articles, government data).

Based on Nature:

↳ Qualitative → Descriptions, opinions, and meanings.

↳ Quantitative → Numerical values and measurements.

8.) Data Collection Tools

Tool	Use
Google Forms	Surveys
Observation Logs	Record behavior
Wireshark	Network packets
Perf Mon	CPU and memory monitoring
Beautiful Soup/Selenium	Web scraping

Choose the tool according to the data required.

9.) Data Analysis and Interpretation

Analysis: Organizing and examining data to find patterns, differences, and relationships.

Interpretation: Explaining what the results mean for the research problem.

Flow: Collect → Analyze → Interpret → Conclude

Remember: A pattern in data does not automatically prove its cause.

10.) Research Instrumentation:
Instrumentation means the hardware and software used for data collection and analysis.

Examples: Sensors; Simulators; Benchmarking tools; Custom AI and network tools.

Calibration: Ensures accurate measurements.

Repeatability: Produces consistent results when an experiment is repeated under the same conditions.

— × — × —

5900123539
KUSHAGRA SINGH
B-1 [CCVT]

__/__/__

Research Methodology — Worksheet-2 [Summary]

1.) Dialectic of Research

Dialectic is structured reasoning using different viewpoints to improve a research claim.

Process: Claim → Challenge → Evidence → Refinement.

It helps identify weakness, consider alternatives, and strengthen research arguments.

2.) Models of Arguement

Type	Meaning	Result
Deduction	General → Specific	Certain if premises are true.
Induction	Specific → General	Probable
Abduction	Observation → Explanation	Plausible

→ Deduction: Applies a known rule to a specific case.

→ Induction: Forms a general rule from observations.

→ Abduction: Finds the most likely explanation for an observation.

3.) Proof Methods

Method	Meaning
Demonstration	Shows functionality through examples / prototypes.
Empirical	Uses experiments and real data.
Mathematical	Uses logic and formal proofs

Empirical process : question → Experiment → Measurement → Analysis → Conclusion

4.) Mathematical Proofs :

↳ Direct Proof : Derive the result from unknown assumptions.

↳ Contradiction : Assume the opposite and derive an impossible result.

↳ Induction : Base case → Hypothesis → Inductive step → conclusion.

↳ Contrapositive : Prove $\neg Q \rightarrow \neg P$ to establish $P \rightarrow Q$.

Induction formula :

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Contrapositive :
Original : $P \rightarrow Q$
Contrapositive : $\neg Q \rightarrow \neg P$

Both are logically equivalent.

5.) Deduction vs Induction vs Abduction

↳ Deduction applies known knowledge
↳ Induction generalizes from observations.
↳ Abduction searches for the best explanation.

[Page 2]

6.) Theoretical Models

→ Theoretical Models are abstract representations of computational systems.
→ Examples : Finite Automata, Pushdown Automata, Turing Machines, Graphs, Boolean Algebra, and Formal Grammars.

Turing Machine : Used to study computability and decidability.

Advantages : Clarity, formal analysis, and hardware-independent reasoning.

Limitation : May ignore real-world constraints.

7.) Algorithmic Approaches

Process:

Problem → Algorithm → Complexity → Correctness → Testing.

Includes Time Complexity, space complexity, correctness, and test cases.

Remember: Successful implementation does not guarantee correctness for every input.

8.) Software Engineering Approaches

Common models: Waterfall, Agile, Spiral.

Cycle: Requirements → Design → Implementation → Testing → Maintenance.

Version control, testing, documentation, and reproductibility improve research software.

9.) Mathematical Modelling

Converts a real-world system into a mathematical representation.

Process:

System → Assumptions → Variables → Model → Analysis → Validation.

Queueing: λ = Arrival rate

μ = Service rate

If $\lambda > \mu$, the queue may grow

10.) Performance Evaluation

Measures system effectiveness

Parameter	Meaning
Time	Execution / response time
Space	Memory usage.
Scalability	#workloads Performance with increased #
Reliability	Consistency and failure behaviour

Evaluation cycle : Metric → Experiment → Measurement → comparison → Interpretation.

ML Metrics : Accuracy, Precision, Recall, F1-score, Runtime :

11.) Performance Tool

↳ Profilers : gprof, Valgrind
↳ Simulation : NS3, CloudSim
↳ Benchmarking : SPEC CPU, ML Perf
↳ Monitoring : Prometheus, Grafana.

× × ×

5900123533
KUSHAGRA SINGH
B-1 [CCVT]

__/__/____

Research Methodology — Worksheet-3 [Summary]

1.) Literature Search:

→ Systematic process of finding existing research on a topic.
→ **Purpose**: Understand previous work, find research gaps, avoid duplication.
→ **Sources**: Research papers, books, conference papers, IEEE, ACM, Scopus, Google Scholar.

2.) Literature Review:

→ Study and evaluation of existing research.
→ **Process**: Question → Keywords → Search → Screen → Analyse → Synthesize.

3.) Keywords and Boolean Operators:

Use relevant and specific keys.

↳ AND → narrows search
↳ OR → broadens search
↳ NOT → excludes terms
↳ " " → exact phrase

4.) Citation Chaining:

→ Backward: Check references of a paper.
→ Forward: Find newer papers citing it.

5.) Critical Analysis:

→ Evaluate a paper based on:
Problem → Method → Dataset → results → limitations

Summary tells what the paper says; analysis evaluates it.

6.) Literature Synthesis & Mapping:

Synthesis : Combine findings from multiple papers.

Research Gap : Unexplored or insufficiently studied area.

Mapping: Organizes research by themes, methods, timelines or citations.

[Page - 2]

7.) Reference & Citations

→ Give credit to resources and prevent plagiarism.

→ Common styles :

↳ IEE : [1], [2]

↳ APA : (Author, Year)

Tools : Zotero, Mendeley, EndNote.

8.) Plagiarism :

→ Using someone else's words, ideas, code or work without proper credit.

Types: Direct, Mosaic, Self-plagiarism, para-phrasing, Accidental.

__/__/__

Paraphrasing: Rewrite in your own words and structure + cite the sour

9.) Research Ethics:

Principles for responsible research:

- Honesty
- Objectivity
- Integrity
- Respect
- Accountability

Fabrication: Making up data
Falsification: Manipulating data/ results.

10.) Data Privacy:

- Obtain consent when required.
- Collect necessary data only.
- Protect personal information
- Anonymize data where appropriate.

11.) Authorship:

→ Authorship should be based on meaningful contribution.

- Gift authorship: Credit without meaningful contribution.
- Ghost authorship: Significant contributors not credited.

12.) IRB:

IRB = Institutional Review Board
Reviews research invoving humans to ensure:

Consent + Privacy + Safety + Minimal Risk
+ Risk to Withdraw.

KUSHAGRA SINGH
590012353
B-1 [CCVT]

_/_/_

Research Methodology — [Mid Sem Paper Summary]

Section - A

(Q1.) Steps: Identify problem → literature review → hypothesis → design methodology → collect data → analyze → interpret → report.

(Q2.) Necessity: gives direction / scope, avoids wasted effort, guides methodology, makes objectives measurable.

(Q3.) Plagiarism = using other's work without credit. Avoid via: citation, para-phrasing, quotation marks, plagiarism checkers, original analysis.

(Q4.) Partly agree — research needs fact-finding / evaluation, but also hypothesis formulation and systematic methodology to generate new knowledge.

(Q5.) Literature review = survey of existing research to find gaps.
Effective approach: define scope → search databases → screen quality → synthesize → identify gaps.

__/__/__

Section – B

Ans 6.) Designs: Randomized Control, Randomized Block, Latin Square, Factorial — each isolates treatment effects from confounds.

Ans 7.) a.) Hypothesis = testable prediction of relationship between variables; purpose is to guide data collection / analysis.

b.) Deductive: theory → specific test.

Inductive → observations → general theory.

Ans 8.) → Fundamental: Knowledge for its own sake, no application.
→ Basic: builds theory, foundation for applied work.
→ Applied: solves specific practical problems.
→ Industrial: applied research for industry product / process improvement.

Ans 9.) Inoculated survival: 15/17 (≈ 88%).
Non-inoculated: 4/14 (≈ 29%).
Inference: vaccine significantly improves survival — effective.

OR: (a) same as Q2.
(b) Modern surveys: online databases, citation tracking, systematic review tools, AI-assisted search.

Section – C

Ans 10.) → Exeperience survey : consult esperienced people to refine problem.

→ Pilot survey : small trial run to test feasibility.

→ Components of research problem : unit/population, variables, context, objectives.

→ Rephrasing : turning vague problems into specific, researchable questions.

~~Ans 11.) a.) Is copying 3 sentences + changing some words + adding citation plagiarism~~

Ans 11.) a.) Yes. Changing a few words but keeping the same sentence structure and ideas is called "patchwriting" — a form of plagiarism. A citation only credits the source ; it doesn't excuse copying its wording. To avoid it, you must fully rewrite in your own words AND cite.

b.) Detection tools scan for matching text strings. Paraphrasing changes the wording / sentence structure, so it won't match word-for-word — lowering the similarity score. But it always stays honest only if you still cite the original source, since the idea still belongs to someone else.

__/__/___

c.) → May still copy the original's structure/flow too closely (still plagiarism, just harder to detect).

→ Can introduce factual errors or change meaning.

→ Author may not fully understand what they're submitting.

→ Raises the question: whose work is it — the AI's, the original author's, or the student's?

d.) Citation credits who the idea/information originally came from. Paraphrasing only changes the words, not the ownership of the idea. So no — paraphrasing alone never removes the need to cite; you always need both.

X X X